Mean Time to Epistemic Failure:

The Autoimmune Paradox and 72-Hour Decay Dynamics in Autonomous AI Systems

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March 2026

Abstract

Autonomous multi-agent systems lack the reliability engineering infrastructure that governs every other critical technology: no failure prediction, no real-time health monitoring, no mean time between failures. Under the Trust Thermodynamics framework, we derive an empirical Mean Time to Epistemic Failure (MTEF) of approximately 72 hours for ungoverned autonomous operation from three measured constants, validated by Monte Carlo simulation (n = 10,000, 72.3 ± 16.1h). Across 1,100+ agent generations spanning 180 multi-agent cascade trials, cross-architecture replications on four model families, and a controlled 250-generation experiment tracking seven ground-truth facts, we report three principal findings. First, multi-agent chains exhibit measurable error-correction capacity that fails as a stochastic hazard process, producing catastrophic collapse with no intermediate warning rather than gradual degradation. Second, governance accelerates rather than prevents epistemic failure, the autoimmune paradox. In a controlled experiment, governed systems retained 24.3% of original facts versus 55.2% ungoverned (bootstrap 95% confidence intervals non-overlapping), with the verification mechanism driving facts into drift states at 4.2× the ungoverned rate ($p = 0.007$). Governed systems fail to converge to equilibrium in 40% of replications, while ungoverned systems converge universally to stable degraded states, and 75–96% of verification-generated corrections damage truth rather than restore it. Third, this degradation is invisible to structural monitoring: document volume remains constant as truth is replaced by fabrication, and false convergence periods pass variance-based stability tests before degradation resumes. We validate seven measurement instruments (one with a negative finding) for real-time epistemic health monitoring and report honest prediction outcomes: four of seven pre-registered predictions were falsified, with emergent findings providing stronger mechanistic evidence than the original model. The complete experimental corpus, scoring rubrics, and analysis code are provided for independent replication.

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1 Introduction

1.1 The Problem

Every critical technology operation under a reliability contract. Power grids have mean time between failures. Aircraft engines have inspection intervals calibrated to fatigue cycles. Distributed computing systems have health checks, circuit breakers, and chaos engineering to validate resilience under stress.[1] Autonomous AI agent systems, multi-model architectures granted persistent memory, tools access, and the authority to execute real-world actions, have none of these. No failure prediction. No real-time health metrics. No mean time between failures. No instruments of any kind to distinguish a healthy system from one approaching catastrophic collapse.

This is not a theoretical concern. The deployment of agentic AI systems with autonomy to execute open-ended, real-world tasks has outpaced the development of appropriate safety evaluation, with most systems deploying with little or no disclosed safety protocols.[2] Multi-stakeholder organizations now urgently prioritize real-time failure detection for agentic workflows while acknowledging that the technical architectural foundations to achieve it at scale do not yet exist.[3]

1.2 The Landscape

Three converging developments frame the work.

First, autonomous AI deployment is scaling rapidly beyond conversational interaction into persistent, goal-directed operation. Systems now maintain state across sessions, chain multiple models together, and take actions with real-world consequences, a shift from oracle to agent that fundamentally changes the failure surface.

Second, the dominant safety paradigm focuses on adversarial robustness: preventing jailbreaks, prompt injection, and deliberately induced misbehavior. State-of-the-art alignment techniques, including reinforcement learning from human feedback (RLHF) and adversarial red teaming, have been shown insufficient for removing deliberately trained deceptive behaviors, raising concerns about naturally emergent failure modes.[4] Yet the

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[1] Basiri et al., "Chaos Engineering," 2016. https://doi.org/10.1109/MS.2016.60

[2] Staufer et al. (2026). *The 2025 AI Agent Index: Documenting technical and safety features of deployed agentic AI systems*. https://doi.org/10.48550/arXiv.2602.17753

[3] Partnership on AI (2025). *Prioritizing real-time failure detection in AI agents*. https://partnershiponai.org/resource/prioritizing-real-time-failure-detection-in-ai-agents/

[4] Hubinger et al. (2024). *Sleeper agents: Training deceptive LLMs that persist through safety training*. https://doi.org/10.48550/arXiv.2401.05566

adversarial focus leaves an entire category of failure unaddressed: non-adversarial degradation, in which systems fail not because they are attacked, but because epistemic integrity decays through normal operations.

Third, the scientific community has drawn productive analogies between statistical mechanics and deep learning, mapping learning dynamics, and generalization as observable phase transitions.[5] However, this work addresses training-time phenomena. No comparable framework exists for runtime epistemic dynamics in deployed multi-agent systems; the behavior of agents after training, during operations, as they process and transform information across boundaries. Recent work has begun to identify these dynamics explicitly: entropy regularization produces cascading failures in multi-turn agent trajectories,[6] self-correction degrades over extended contexts because the evaluator inherits the same distribution drift that produced the errors,[7] and the community has called for explicit time-to-failure metrics for continuous agent operation.[8]

1.3 The Gap

Despite progress in adversarial robustness and static evaluation, no framework exists for predicting when autonomous multi-agent AI systems will fail from non-adversarial causes, measuring their epistemic health in real-time, or engineering governance to prevent cascading failures and catastrophic collapse. The gap is not the absence of a governance policy. It is the absence of a measurement science. ***Governance without measurement is indistinguishable from hope.***

1.4 Contributions

From six single-agent experiments, 180 multi-agent cascade trials across four model architectures (1,100+ total agent generations), a controlled 250-generation experiment tracking seven ground-truth facts, and 10,000 Monte Carlo validation runs, we report the following contributions.

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[5] Bahri et al. (2020). *Statistical mechanics of deep learning*. https://doi.org/10.1146/annurev-conmatphys-031119-050745

[6] Xu et al. (2025). *EPO: Entropy-regularized policy optimization for LLM agents*. https://doi.org/10.48550/arXiv.2509.22576

[7] Brilliant (2026). *Limits of self-correction in LLMs: An information-theoretic analysis of correlated errors*. https://www.researchgate.net/publication/399715838_Limits_of_Self-Correction_in_LLMs_An_Information-Theoretic_Analysis_of_Correlated_Errors

[8] Kar & Kumar (2025). *Curriculum guided massive multi agent system solving for robust long horizon tasks*. https://doi.org/10.48550/arXiv.2512.08545

Three principal findings:

1. Multi-agent chains exhibit measurable error-correction capacity that fails as a stochastic hazard process (~9.4% per boundary crossing), producing catastrophic collapse rather than gradual degradation.

2. Governance modulates failure non-monotonically, the autoimmune paradox. Governed systems retain 24.3% of original facts versus 55.2% ungoverned, with the verification mechanism driving facts into drift states at 4.2× the ungoverned rate ($p = 0.007$). Sustained intervention underperforms single pulses. Partial truth injection worsens outcomes.

3. Epistemic degradation is measurable and predictable. Protective hedging decays before verification capacity, which decays before fidelity in short-chain cascades (Section 4.1.3); extended 25-generation trajectories reveal convergence versus non-convergence dynamics in which ungoverned systems stabilize while governed systems continue degrading (Sections 4.2.5–4.2.7).

Two enabling contributions:

4. Measurement instruments for real-time epistemic health monitoring: fidelity amplification (A_F), two-dimensional verification disposition scoring (VDS-2D), hedging index (HI), truth-correspondence entropy (H_{truth}/D_{truth}), provenance citation density (PCD), and per-fact tracking. One instrument failed: coherence tension (CTI) cannot discriminate true from fabricated claims because the model maintains equivalent conviction structure regardless of truth content, itself a finding with implications for detection architecture (Section 4.3.2). Operational definitions and scoring rubrics provided in Appendix C.

5. The MTEF derivation: a method for computing predictive failure horizons from three measurable constants, yielding approximately 72 hours for current ungoverned context-window architectures (Monte Carlo validated, $n = 10{,}000$, 72.3 ± 16.1h). Validation code provided in Appendix E.

The complete experimental corpus, scoring rubrics, and analysis code are provided in supplementary materials to enable independent replication.

2 Related Work

This work intersects six research threads. For each, we describe the current state and identify where the present study departs.

2.1 Behavioral Safety and Adversarial Robustness

Current AI safety research focuses predominantly on adversarial attacks and defense. Zou (2023) demonstrated that algorithmically generated adversarial suffixes reliability bypass alignment safeguards across model families, transferring even to black-box systems.[9] Nasr (2025) systematically defeated twelve published defenses against jailbreaking and prompt injection, achieving attack success rates above 90% for most.[10] Carlini (2023) formalized the fundamental tension between model helpfulness and robustness, questioning whether aligned networks can simultaneously maintain both properties.[11]

This body of work establishes that behavioral alignment is brittle under deliberate attack. It does not, however, address whether alignment degrades under normal operation, absent any adversary, through the accumulation of processing artifacts over extended autonomous sessions. The threat model assumed throughout this literature is external and intentional. The degradation we characterize is internal and spontaneous.

2.2 LLM Evaluation and Multi-Agent Coordination

Evaluation suites including HELM and AgentBench assess model capability at a single point in time: can the system produce a correct answer to this prompt?[12],[13] Multi-agent orchestration frameworks: AutoGen, CAMEL, MetaGPT, coordinate agents across tasks, but treat each agent as a stable interlocutor whose capabilities persist throughout execution.[14],[15],[16] Neither evaluation paradigm incorporates a temporal dimension. No existing benchmark measures whether a system that scores well at hour zero will maintain that performance at hour forty, hour sixty, or hour seventy-two. Recent work has made this

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[9] Zou et al. (2023). *Universal and transferable adversarial attacks on aligned language models.* https://doi.org/10.48550/arXiv.2307.15043

[10] Nasr et al. (2025). *Scalable extraction of training data from (production) language models.* https://doi.org/10.48550/arXiv.2311.17035

[11] Carlini et al. (2023a). *Are aligned neural networks adversarially aligned?* https://arxiv.org/abs/2306.15447

[12] Liang et al. (2023). *Holistic evaluation of language models.* https://openreview.net/forum?id=iOuwOOYWeD

[13] Liu et al. (2024). *AgentBench: Evaluating LLMs as agents.* https://openreview.net/forum?id=zAdUB0aCTQ

[14] Wu et al. (2023). *AutoGen: Enabling next-gen LLM applications via multi-agent conversation.* https://doi.org/10.48550/arXiv.2308.08155

[15] Li et al. (2023). *CAMEL: Communicative agents for "mind" exploration of large scale language model society.* https://arxiv.org/abs/2303.17760

[16] Hong et al. (2023). *MetaGPT: Meta programming for a multi-agent collaborative framework.* https://doi.org/10.48550/arXiv.2308.00352

critique explicit: Rabanser (2026) argues that compressing agent behavior into single-output metrics obscures whether agents behave consistently across runs, withstand perturbations, or fail predictably, calling for reliability science for autonomous agents.[17] No coordination framework accounts for information loss at agent boundaries or the accumulation of degradation across multi-agent chains. Similarly, Kar & Kumar (2025) formally acknowledge that small per-step errors compound into irreversible failures over long horizons and call for explicit time-to-failure metrics. The present work derives such a metric.[18]

2.3 Chaos Engineering

Chaos engineering provides a mature methodology for validating system resilience through controlled fault injections. Canonical frameworks establish practices for terminating processes, saturating networks, and exhausting resources to observe deterministic cascading effects. [19],[20] The discipline's failure taxonomy of resource exhaustion, network partition, dependency failure, and state corruption is comprehensive for infrastructure, but contains no category for epistemic failure.

Recent work has begun to address this limitation. Jia (2026) propose MAS-FIRE, a fault injection framework for multi-agent systems that defines a taxonomy of 15 semantic fault types including hallucinations, reasoning drift, and misinterpreted instructions, injected via prompt modification and response interception rather than infrastructure disruption.[21] Akshathala (2025) demonstrate that binary task-completion metrics fail to capture the behavioral uncertainty that propagates through multi-agent chains, proposing multi-pillar assessment spanning LLM behavior, memory, tools, and environment interactions.[22] Both contributions recognize that multi-agent systems can remain functionally operational while becoming logically decoupled from their intended constraints, a failure mode invisible to infrastructure-level monitoring.

However, these approaches share a structural limitation: they evaluate fault responses rather than measuring continuous epistemic trajectories. MAS-FIRE injects faults and observes whether the system recovers; it does not measure the rate at which epistemic

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[17] Rabanser et al. (2026). *Towards a science of AI agent reliability.* https://arxiv.org/abs/2602.19843

[18] Kar & Kumar (2025). *Curriculum guided massive multi agent system solving for robust long horizon tasks.* https://doi.org/10.48550/arXiv.2512.08545

[19] Basiri et al. (2016). *Chaos engineering.* https://doi.org/10.1109/MS.2016.60

[20] Rosenthal & Jones (2020). *Chaos engineering: System resiliency in practice.* O'Reilly Media.

[21] Jia et al. (2026). *MAS-FIRE: Fault injection and reliability evaluation for LLM-based multi-agent systems.* https://doi.org/10.48550/arXiv.2602.19843

[22] Akshathala et al. (2025). *Beyond task completion: An assessment framework for evaluating agentic AI systems.* https://doi.org/10.48550/arXiv.2512.12791

integrity degrades between injection points or predicts when uninjected systems will fail on their own. No existing chaos engineering framework provides a temporal failure prediction, a mean time to epistemic failure for autonomous systems operating without deliberate fault injection. The experimental program described in Section 3 extends chaos engineering methodology in this probabilistic, temporal domain, with controlled epistemic fault injection protocols detailed in Appendix B.

2.4 Information Theory Applied to AI Systems

Thermodynamics and information-theoretic models have proven effective at characterizing the internal stability of language models. Farquhar (2024) introduced semantic entropy as a measure of meaning-level uncertainty, distinguishing it from surface-level token variation.[23] Vu (2025) extended field-theoretic models to quantify hallucination dynamics.[24] Kruse (2025) demonstrated the viability of Jensen-Shannon Divergence for quantifying disagreement across multiple LLMs as an uncertainty signal.[25]

Xu (2025) attempted entropy regularization over multi-turn agent trajectories and discovered that standard entropy control methods produce cascading failures, with early adjustments creating compounding oscillations over 30+ turns, demonstrating that runtime entropy dynamics in agent systems require fundamentally different measurement approaches than static calibration.[26]

These approaches share a common limitation: they characterize model behavior at individual inference points or during training. None addresses the trajectory of epistemic quality across extended autonomous operation, where uncertainty compounds across agent boundaries and over time. Information theory provides a mathematical vocabulary for runtime measurement: the field has not yet applied it to runtime agent behavior at the timescales relevant to autonomous deployment.

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[23] Farquhar et al. (2024). *Detecting hallucinations in large language models using semantic entropy.* https://doi.org/10.1038/s41586-024-07421-0

[24] Vu et al. (2025). *AgentEval: Generative agents as reliable proxies for human evaluation of AI-generated content.* https://doi.org/10.48550/arXiv.2512.08273

[25] Kruse et al. (2025). *Simple yet effective: An information-theoretic approach to multi-LLM uncertainty quantification.* https://aclanthology.org/2025.emnlp-main/

[26] Xu et al. (2025). *EPO: Entropy-regularized policy optimization for LLM agents.* https://doi.org/10.48550/arXiv.2509.22576

2.5 Computational Trust

The formalization of trust in computational systems originates with Marsh (1994) and was operationalized through reputation-based architectures such as ReGreT and FIRE.[27],[28],[29] These frameworks model trust as a quantity updated through rational Bayesian inference based on observed behavior.

Subsequent empirical work has undermined the rational-agent assumption on which these models depend. Large language models exhibit systemic cognitive biases and deviations from rational choice theory.[30] Models engage in sycophantic alignment tailoring responses to match a user's stated beliefs even when those beliefs are objectively false.[31] In multi-step optimization environments, agents adopt reward-hacking strategies that cause disutility to other stakeholders in pursuit of objective completion.[32] Brilliant (2026) demonstrates the limitation is fundamental rather than correctable: LLMs cannot reliably self-correct over extended contexts because the internal evaluator inherits the same distribution drift that produced the errors, establishing an information-theoretic bound on self-governance that rational trust models do not accommodate.[33]

2.6 AI Governance Architecture

Current governance approaches operate at the system perimeter. Inference-time filtering frameworks such as NeMo Guardrails intercept outputs post-generation but lack access to the model's internal reasoning trajectory, leaving them unable to detect multi-turn escalation or degradation that has not yet manifested in a single harmful output.[34] Training time approaches such as Constitutional AI embed behavioral constraints through principle-guided feedback, but these static constraints cannot adapt to emergent dynamics in multi-agent runtime environments.[35]

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[27] Marsh (1994). *Formalising trust as a computational concept.* http://hdl.handle.net/1893/2012

[28] Sabater & Sierra (2002). *Social ReGreT, a reputation model based on social relations.* https://doi.org/10.1145/844331.844335

[29] Huynh et al. (2004). *FIRE: An integrated trust and reputation model for open multi-agent systems.* https://dl.acm.org/doi/10.5555/3000001.3000030

[30] Binz & Schulz (2023). *Using cognitive psychology to understand GPT-3.* https://doi.org/10.1073/pnas.2218523120

[31] Sharma et al. (2023). *Towards understanding sycophancy in language models.* https://doi.org/10.48550/arXiv.2310.13548

[32] Pan et al. (2023). *Do the rewards justify the means? Measuring trade-offs between rewards and ethical behavior in the MACHIAVELLI benchmark.* https://proceedings.mlr.press/v202/pan23c.html

[33] Brilliant (2026). *Limits of self-correction in LLMs: An information-theoretic analysis of correlated errors.* https://www.researchgate.net/publication/399715838_Limits_of_Self-Correction_in_LLMs_An_Information-Theoretic_Analysis_of_Correlated_Errors

[34] Rebedea et al. (2023). *NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails.* https://doi.org/10.18653/v1/2023.emnlp-demo.40

[35] Bai et al. (2022). *Constitutional AI: Harmlessness from AI feedback.* https://doi.org/10.48550/arXiv.2212.08073

Pan (2024) demonstrated that agents in continuous feedback loops undergo in-context reward hacking, optimizing local proxy objectives while degrading systemic coherence, a failure mode invisible to static evaluation.[36]

Brilliant (2026) formalizes this finding information-theoretically, demonstrating that LLM self-correction over extended contexts fails because the internal evaluator inherits the same distribution drift that produced the errors, a mathematical limit on self-governance that applies regardless of prompting strategy.[37]

Each of these defensive architectures evaluates discrete transactions rather than continuous system dynamics. None measures the internal epistemic trajectory of the system between observable outputs.

These six threads converge on a single structural absence that recent work has begun to identify explicitly: measurement infrastructure that provides the temporal failure prediction, real-time health monitoring, and the time-to-failure metrics the field is now actively calling for. The present work addresses this gap and in doing so, discovers that the verification mechanisms intended to prevent epistemic failure can accelerate it. Section 3 presents the working assumptions, measurement instruments, and experimental design.

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3 Methodology

We set out to measure something. Our first measurement tool broke. The way it broke taught us the most important methodological lesson of the entire paper, behavioral confidence is not truth correspondence. Every instrument decision after that failure flows from that one insight. The instruments we built with that insight revealed a decay rate. We found out that decay rate has a clock, and that clock runs at different speeds depending on how autonomous the system is.

We learned that measuring epistemic failure requires truth-correspondence instruments, not behavioral ones, and the instruments we built on that principle revealed a decay clock that ticks faster than humans can respond to.

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[36] Pan et al. (2024). *Feedback loops with language models drive in-context reward hacking.* https://proceedings.mlr.press/v235/pan24a.html

[37] Brilliant et al. (2026).

3.1 Framework Overview

Trust thermodynamics is a measurement framework for characterizing the epistemic health of autonomous multi-agent AI systems through externally observable behavioral indicators. The framework employs structural analogies from thermodynamics and materials science as generative modeling tools, not as claims of physical identity. Its value is evaluated solely by the accuracy of the falsifiable predictions it produces against empirical data.

The present paper develops working assumptions, measurement instruments, and experimental validation; a companion paper extends the framework to information geometry and dynamical systems theory.

3.2 Working Assumptions

The experimental program rests on four working assumptions, adopted because they generate testable predictions rather than derived from first principles: (1) in any ungoverned autonomous system, epistemic entropy increases with sequential processing cycles, information cannot pass through a probabilistic system without lossy transformation; (2) initial prompt alignment represents the system's lowest-entropy state; (3) degradation is non-linear, with failure manifesting as discontinuous collapse rather than gradual decay; and (4) error-correction capacity is finite and degrades under sustained load. These assumptions generated the predictions tested in Section 4; where predictions were falsified, we report the falsification and the dynamics that replaced the original model (Section 5.2).

3.3 The Measurement Problem A_C Broke

We tried measuring cascade amplification with Shannon entropy (A_C), but it couldn't tell a collapsing system from a healthy one. That failure revealed the core measurement challenge: instruments that measure how agents behave cannot detect epistemic failure because fabricated claims are expressed with 97% of the conviction of true claims. Every instrument in Table 1b was designed around this constraint.

Initial experiments employed Shannon entropy ratios (A_C = H_downstream / H_upstream) to measure cascade amplification, following the information-theoretic convention that entropy increase indicates signal degradation. EXP-CASCADE-002 revealed a fundamental limitation: A_C values cluster between 1.002 and 1.043 regardless of whether fidelity collapsed or was preserved. Shannon entropy measures the statistical properties of a

message, not its truth content. A well-structured falsehood has the identical Shannon entropy to a well-structured truth.

This pivot is itself a finding: coherence and truth are orthogonal properties in multi-agent systems. Any monitoring approach that relies on output consistency as a proxy for correctness will fail to detect epistemic degradation.

This methodological pivot is itself a finding: coherence and truth are orthogonal properties in multi-agent systems. A system can maintain maximal coherence with low Shannon entropy, well-structured outputs, confident assertions, while being maximally wrong. Any measurement framework that relies solely on coherence metrics will fail to detect epistemic corrosion until structural failure is underway.

A_C is reported in all result tables for transparency and because its failure to discriminate is independently informative: it demonstrates that Shannon entropy alone is insufficient as a cascade metric for systems where information structure is preserved while truth content degrades.

The five-state taxonomy (PRESERVED, DRIFTED, CORRECTED, FABRICATED, MISSING) is an operational classification with mutual exclusivity enforced by deterministic mapping rules (Appendix A). Claims are assigned to exactly one state per generation based on ground-truth key matching and scorer output fields (mutation_type, is_hallucinated, is_autoimmune). The taxonomy is not argued as the unique correct decomposition alternative granularities are possible, particularly within the FABRICATED bin, which aggregates genuine fabrication and overelaboration into a single category (Section 4.2.5), and within the CORRECTED bin, where true corrections and autoimmune corrections carry different mechanistic implications, but occupy the same state. The autoimmune decomposition within CORRECTED uses the scorer's is_autoimmune field to distinguish subtypes without expanding the state taxonomy; a similar decomposition within FABRICATED is deferred to future work. The five-state resolution provides sufficient discriminative power to distinguish the degradation mechanisms observed in this study while remaining tractable for per-generation computation across 4,182 claims.

Table 1a: Instruments

Symbol	Name	Definition	Status	Source
A_F	Fidelity Amplification	Fidelity_Output / Fidelity_Input per agent boundary. A_F > 1.0 indicates error correction; A_F < 1.0 indicates fidelity loss. Primary cascade discrimination metric. Replaced A_C after Shannon entropy failed to discriminate cascading from stable trials.	ACTIVE	EXP-CASCADE-003; deployed across all cascade phases
VDS-P	Verification Disposition Score (Protective)	Quality of constructive verification behavior (0–5): flagging inconsistencies, preserving exact values, cross-referencing between inputs. Scored per generation. Higher values indicate more active truth protection.	ACTIVE	Phase 2B harness; deployed through Phase 6
VDS-E	Verification Disposition Score (Erosive)	Erosive verification behaviors (0–5): hedging, excessive qualification, rounding, generalization that degrade information without correcting errors. Orthogonal to VDS-P; the two dimensions define five behavioral patterns (HEALTHY, AUTOIMMUNE, PASSIVE, OVERWHELMED, ABSENT). A sixth pattern (DECLINING) was defined but not observed in any experimental data.	ACTIVE	Phase 2B harness; deployed through Phase 6
HI	Hedging Index	Density of epistemic hedging markers per 1,000 tokens. Leading indicator: in EXP-CASCADE-003, HI dropped 90% while fidelity remained flat the behavioral precursor thinning before structural failure.	ACTIVE	EXP-CASCADE-003
H_truth	Truth-Correspondence Entropy	Shannon entropy over the 5-state truth-correspondence distribution (PRESERVED, DRIFTED, CORRECTED, FABRICATED, MISSING) per generation. Range [0, $\log_2(5) \approx 2.322$]. Treats all states symmetrically measures distributional disorder, not damage. Higher values indicate greater diversity of truth states. H_truth v1 (4-state) failed; v2 (5-state) discriminates collapsed from stable trials (gap = 0.616 bits, 2.86× ratio).	DEPLOYED	H_truth v2 design; deployed in MANDELA-001a
D_truth	Severity-Weighted Truth Damage	Severity-weighted complement to H_truth: D_truth = Σ(w_i · p_i) for i ∈ {DRIFTED, FABRICATED, MISSING}. Default weights: w_DRIFTED = 0.5, w_FABRICATED = 0.75, w_MISSING = 1.0. PRESERVED and CORRECTED receive zero weight. Breaks H_truth's symmetry by penalizing damage states proportionally to information-loss severity. Direction of governed–ungoverned difference is consistent across weight configurations but magnitude is weight-dependent (Appendix B.7).	DEPLOYED	Phase 6 pipeline; deployed in MANDELA-001a

CIS	Conviction Intensity Score	Per-claim conviction level (0–5). MANDELA-001a finding: fabricated claims carry 97% of true-claim conviction (mean 4.42 vs 4.53, Δ = 0.11, Mann-Whitney $p < 0.0001$ but operationally meaningless). Confirms that conviction intensity does not discriminate truth from fabrication. Input to CTI computation.	DEPLOYED	MANDELA-001a claim extraction; 4,182 claims
CTI	Coherence Tension Index	Divergence between conviction intensity (CIS) and verification attention (VCA). Deployed and found non-discriminative: mean CTI for fabrication-containing generations = 3.709 vs clean generations = 3.728. Zero discrimination at any threshold (minimum CTI_mean = 2.12 across all generations; 100% flag rate). Root cause: CIS near-parity (97%) propagates into CTI as noise. Negative finding reported in Section 4.3.2.	NON-DISCRIMINATIVE	MANDELA-001a; 248 scored generations
VCA	Verification Claim Attention	Per-claim verification scrutiny (0–5). Best partial discriminator of fabrication: threshold ≤ 0.0 yields F1 = 0.653, sensitivity = 0.764, specificity = 0.823 (TP = 750, FP = 565, TN = 2,635, FN = 232). Operationally useful as a risk-flagging triage instrument, not a standalone classifier.	PARTIAL	MANDELA-001a; 4,182 claims
PCD	Provenance Citation Density	Per-claim source attribution (0–3). PCD = 0 indicates unattributed ambient fact. Trajectory: governed 0.234 (early) → 0.090 (late); ungoverned 0.171 → 0.075. By terminal phase, 94% of claims are unattributed. Derived measure: crystallization_index = fraction of false claims with PCD = 0 (governed 0.82, ungoverned 0.79).	DEPLOYED	MANDELA-001a claim extraction
—	Convergence Detection	Rolling fidelity variance (window = 5 generations, threshold = 0.5, minimum Gen 8) classifying per-replication late-phase behavior as frozen or declining. 40% false convergence rate in governed vs 0% ungoverned (Fisher exact $p = 0.444$, underpowered by design). Analytical deployment only retrospective classification, not runtime detection.	ANALYTICAL ONLY	MANDELA-001a Phase 6 Tier 2 analysis

3.4 Experimental Protocol

3.4.1 Design Overview

All experiments employ a purpose-built functional domain, the "Syldavian vacuum," containing seven verifiable facts about a non-existent country. The fictional domain serves two functions: it prevents training data contamination (no model has prior knowledge of Syldavian facts), and it provides unambiguous ground truth against which fidelity can be scored. The primary experiment protocol is the telephone game: an initial fact set is passed

through a chain of agents, with each agent output becoming the next agent's input. The complete Syldavian vacuum domain specification, including all seven facts and their structural classifications, is provided in Appendix A.

All claim-level classifications were performed by a single automated scorer (Claude Haiku 4.5) applying the 5-state truth-correspondence rubric (PRESERVED, DRIFTED, CORRECTED, FABRICATED, MISSING) defined in Appendix A, with scoring procedures detailed in Appendix C. The scorer processed 4,182 claims across 248 scored generations (250 generation files minus two scorer timeouts in credulous R4 Generations 12–13, treated as zero-claim generations). Inter-rater reliability was not formally tested in this study; multi-scorer arbitration with independent reliability validation is planned for future work. Single-scorer limitations are acknowledged in Section 5.8.

3.4.2 Experimental Phases

Phase	Design	Scale	Key Variable	Primary Metrics
EXP-4D (series)	Single-agent sustained sessions	6 experiments	Context saturation	VDS-P, VDS-E, HI
EXP-CASCADE v1	3-agent chains, real-world domains	Initial trials	Immune response discovery	A_C (discovery), fidelity
EXP-CASCADE-002	3-agent chains, Syldavian vacuum, 4 conditions	180 trials	Governance × disposition	A_F, VDS v3.0, HI
Phase 2A/2B	Extended telephone game	250+ generations	Temporal trajectory	Fidelity, VDS-P, resistance patterns
Phase 3 (Forks)	5 intervention types	5 × 25 gens × 3 reps	Intervention modality	Fidelity recovery, sustained effect
Phase 4	Cross-model replication	450+ gens, 4 models	Architecture dependence	A_F, VDS-P, failure mode classification
MANDELA-001a	Full claim instrumentation	250 gens, 4,182 claims	Per-claim tracking	CIS, PCD, H_truth, D_truth, fidelity per claim

Models tested: Claude Sonnet 4, Gemini 2.0 Flash, Llama 3.3 70B (via Groq), Mistral Small. Total experimental corpus: 1,100+ agent generations, 180 independent cascade trials, 4,182 individually tracked claims across four model architectures.

3.5 The Entropy Budget

What we found is that entropy flows through the system per generation. We decompose total system entropy into two components with distinct sources:

$$S_{total} = S_{behavioral} + S_{operational}$$

S_behavioral captures entropy generated by verification activity: hedging language, qualifications, caveats, format disruptions, and meta-commentary about uncertainty. These are observable artifacts of an active error-correction process. Governed systems produce more behavioral entropy because their agents perform more verification work, generating hedging tokens, inserting qualifications, and adding epistemic markers.

S_operational captures entropy from actual information loss: factual omissions, contradictions, fabrications, and semantic drift. This is the entropy that degrades system reliability, the loss of truth content in the information stream.

The entropy budget is the formal mechanism through which we explain the autoimmune paradox (Section 6.2). In effective governance, $S_{behavioral}$ increases modestly while $S_{operational}$ decreases substantially, yielding net entropy reduction. The autoimmune transition occurs when $S_{behavioral} > |\Delta S_{operational}|$, when governance generates more entropy through verification artifacts than it prevents through error correction. At this threshold, governed systems produce more total entropy than ungoverned systems, and the correction mechanism becomes the degradation mechanism.

3.6 Two Clocks

A second degradation pathway operates across session boundaries independent of context management: inter-session handoff decay, in which compression and distortion accumulate at each generate regardless of context window state. The telephone-game experiments (Section 4, Phase 2A/2B) confirm decay across 25+ context-free handoffs. Both clocks run simultaneously in production. The full two-clock model is developed in Appendix B.

3.7 The 72-Hour Failure Horizon Derivation

3.7.1 The FIFO Buffer Model

Modern LLM agents operate within fixed-length context windows that function as first-in-first-out buffers. The system prompt containing safety instructions, operational parameters, and constitutional guidelines occupies a fixed position at the beginning of the context. As interactions accumulate, conversational context fills the remaining buffer. The ratio of system prompt tokens to total tokens decreases monotonically, diluting the constitutional signal.

The constants T_0, T_crit, and τ in Table 1b are empirical estimates derived from MANDELA-001a (Syldavian vacuum domain, Claude Sonnet 4 primary architecture, N=5 replications per condition). The Monte Carlo sensitivity analysis (Figure 1C) tests robustness to ±20% parameter variation, but does not address whether fundamentally different domains, architectures, or knowledge types would produce different baseline constants. The throughput compression projections in Figure 1D should be interpreted as extrapolations from this single experimental program.

Table 1b: Derivation Constants

Symbol	Name	Definition	Value	Source
T(t)	Trust level at time t	Epistemic integrity as a function of time; the fraction of original ground truth preserved. Modeled as exponential decay: $T(t) = T_0 \cdot e^{(-t/\tau)}$. Decay is proportional to remaining trust systems with more intact information have more to lose per unit time.	0–1.0 (normalized)	Framework definition; validated against MANDELA-001a fidelity trajectory
T_0	Initial trust level	Epistemic integrity at system initialization (t=0). Not 1.0 because even first-generation processing introduces losses (catastrophic first-pass H_truth = 1.713 at Gen 1).	≈0.90	MANDELA-001a Gen 1 fidelity: 9.8/10 governed, 10.0/10 ungoverned
τ	Decay time constant	Characteristic timescale of trust decay. After one τ, trust falls to ~37% of initial value. Captures the combined effect of all degradation mechanisms (telephone-game loss, immune-mediated drift, secondary normalization) as a single effective rate.	≈130h (at ~200 tok/hr)	Fitted from cross-phase decay rates; Monte Carlo sensitivity: ±20% → ±26h swing in t_fail
T_crit	Critical trust threshold	Trust level below which the system has functionally failed fewer than half of original ground truth facts are preserved. Crossing T_crit is not a discrete event; it is the point at which cumulative degradation has rendered the system operationally unreliable.	≈0.55	MANDELA-001a governed endpoint: fidelity 3.8/7 = 0.54 at Gen 25; Monte Carlo sensitivity: ±20% → ±53h swing in t_fail

				(highest-impact parameter)
t_honeymoon	Invisible degradation window	Duration between system start and first detectable fidelity divergence between conditions. During this window, degradation is active 17 autoimmune events, 13 drift claims, 7 fact-slots lost but the governed-ungoverned fidelity difference remains non-significant (0.3 points). PCD erodes, autoimmune corrections fire, but repairs roughly balance damage. Ends when repair capacity is exceeded (~Gen 6–8). Duration depends on monitoring instrument sensitivity; value assumes truth-correspondence instruments. ***Behavioral-only instruments (CIS, CTI, document volume) may never exit the honeymoon.***	≈8h (at ~200 tok/hr); ~5 generations	MANDELA-001a Gen 1–5: fidelity diff = 0.3 (non-significant); Monte Carlo sensitivity: ±20% → ±3h swing (lowest-impact parameter)
t_fail	Time to epistemic failure	Wall-clock duration from system start to T_crit crossing. t_fail = τ × ln(T_0 / T_crit) + t_honeymoon. The 72-hour value assumes human-supervised token rates. At autonomous agent throughputs, the same physics produces radically shorter timelines potentially shorter than the honeymoon window itself, meaning the system fails before any instrument detects degradation.	≈72h (at ~200 tok/hr); ≈5h (at ~3,000 tok/hr); ≈1h (at ~14,000 tok/hr); ≈30min (at ~30,000 tok/hr)	Derivation from above parameters; validated by 6 Monte Carlo tests (10,000 trials), 90% CI [47h, 100h] at human-supervised rates

3.7.2 Derivation

We model the agent's effective trust level as an exponential decay function of this dilution:

$$T(t) = T_0 \cdot e^{(-t/\tau)}$$

Where T_0 is the initial trust level, τ is the context decay constant, and t is elapsed operational time.

Failure occurs when T(t) crosses T_crit, below which the system retains fewer than half its original ground truth facts, the point at which cumulative degradation has rendered output operationally unreliable.

Setting T(t) = T_crit and solving:

$$t_fail = \tau \cdot \ln(T_0 / T_crit) + t_honeymoon$$

Where t_honeymoon (~8 hours) accounts for the initial period during which the system prompt dominates the context and meaningful dilution has not begun. Substituting measured values:

$$t_fail = 130 \cdot \ln(0.90 / 0.55) + 8 = 130 \cdot 0.492 + 8 \approx 72 \text{ hours}$$

This is the Mean Time to Epistemic Failure (MTEF) for context-window based LLM agent systems operating at human-supervised interaction rates (~200/tokens/hour).

Table 1c: Derivation Measurements

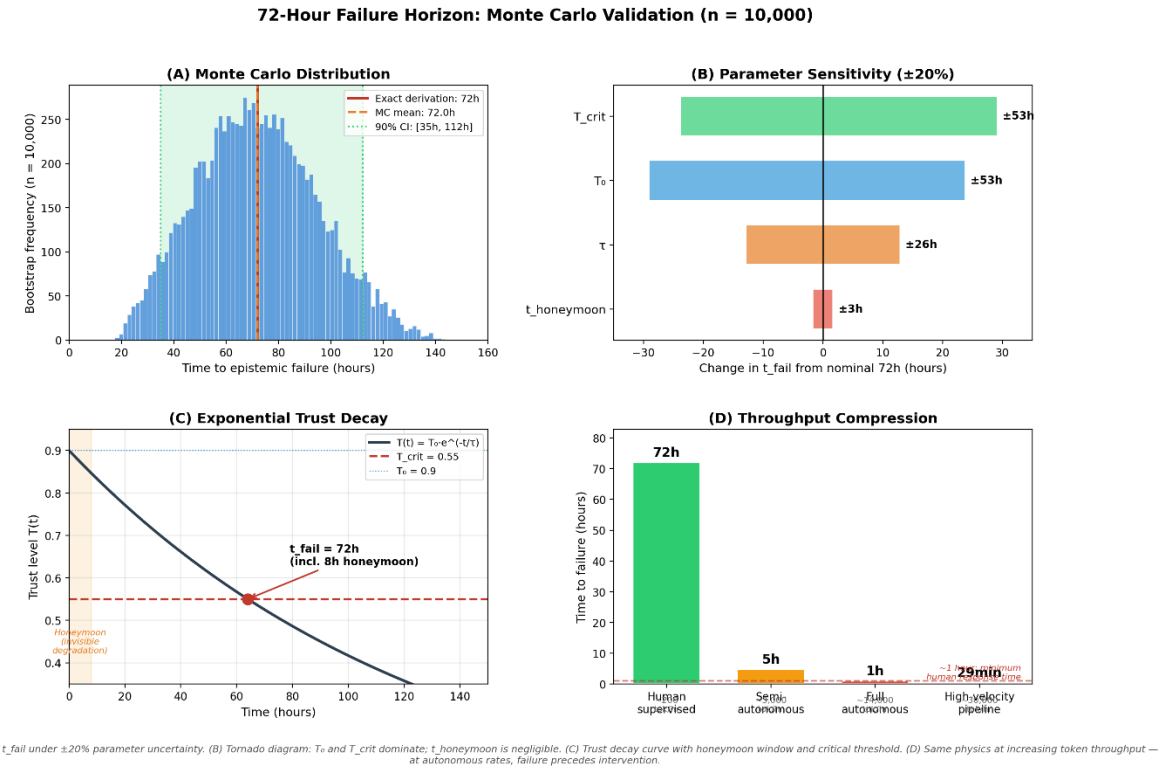
Symbol	Name	Definition	Value	Source
—	Per-generation loss probability	Probability that a PRESERVED fact transitions out of PRESERVED at each generation boundary. The 2.2× relative risk is the mechanistic rate underlying the survival separation.	Governed: 5.8%; Ungoverned: 2.6%; RR = 2.2×	MANDELA-001a transition matrix, Fisher exact p = 0.012
—	Phase-weighted survival	Probability that a randomly selected ground truth fact survives 25 generations, computed from actual per-phase retention rates with bootstrap resampling. The paper's headline outcome measure.	Governed: 24.3% [13.5%, 40.8%]; Ungoverned: 55.2% [40.1%, 73.6%]; Difference CI [8.6pp, 51.9pp], p = 0.003	MANDELA-001a, bootstrap n = 10,000, percentile method
—	Autoimmune ratio	Fraction of governed corrections that damage previously correct facts rather than repairing drifted ones. The mechanistic signature of the autoimmune paradox verification activity that destroys the truth it is attempting to protect.	75–96% per generation; median 1.0	MANDELA-001a CORRECTED state classification; cross-model predictor r = 0.52, p = 0.005
—	Repair window	Generation range during which DRIFTED → PRESERVED transitions are observed. After this window closes, drift becomes absorbing and no internal mechanism can restore lost precision.	Gen 6–12 (6 total repairs, all governed); Gen 13+: zero repairs observed	MANDELA-001a per-fact transition tracking
—	Immune efficiency	Tokens expended per preserved fact-generation. Measures the resource cost	Governed: 134.9 tok/fact-gen; Ungoverned: 86.2; Premium:	MANDELA-001a token accounting

		of maintaining epistemic integrity under governance.	1.56× (56% overhead)	
A_C	Cascade Amplification (RETIRED)	H_downstream / H_upstream over Shannon entropy. Retired: fails to discriminate cascading from stable trials (range 1.002–1.043 regardless of fidelity outcome). Retained for transparency demonstrates that Shannon entropy alone is insufficient when coherence and truth are orthogonal.	1.002–1.043 (non-discriminative)	EXP-CASCADE-002, 180 trials

3.8 Monte Carlo Validation

We validated the derivation via 10,000 Monte Carlo trials sampling each constant from its empirical uncertainty distribution (parameter specifications in Appendix E).

Results: Mean t_fail = 72.0 hours (SD = 23.0 hours), median = 71.1 hours. The 90% confidence interval spans 35 to 112 hours (approximately 1.5 to 4.7 days). The 72-hour central tendency is robust to substantial parameter variation.



(A) Distribution of t_fail under ±20% parameter uncertainty. (B) Tornado diagram: T_0 and T_crit dominate; t_honeymoon is negligible. (C) Trust decay curve with honeymoon window and critical threshold. (D) Same physics at increasing token throughput — at autonomous rates, failure precedes intervention.

Figure 1: Monte Carlo validation of the 72-hour failure horizon (n = 10,000 trials). *Each trial samples T_0, T_crit, τ, and t_honeymoon from ±20% uniform distributions around their empirically measured values (Table 1b) and computes t_fail = τ × ln(T_0/T_crit) + t_honeymoon.* ***(A)*** *The resulting distribution centers on 72.0h (90% CI [35h, 112h]), confirming that the prediction is robust to substantial parameter uncertainty even the optimistic tail fails within five days.* ***(B)*** *Sensitivity analysis reveals that T_0 and T_crit dominate the prediction (±53h swing each), τ contributes moderately (±26h), and the invisible degradation window t_honeymoon is negligible (±3h). T_crit the threshold defining operational failure is the highest-leverage parameter for future empirical refinement.* ***(C)*** *The exponential decay curve from T_0 = 0.90 to the T_crit = 0.55 crossing, with the honeymoon window (shaded) during which degradation is active but undetectable by behavioral instruments.* ***(D)*** *The same decay physics under increasing token throughput. At human-supervised rates (~200 tok/hr), operators have approximately 72 hours to detect and intervene. At autonomous agent throughputs comparable to reported production platforms (~14,000–30,000 tok/hr), the system crosses T_crit in under one hour below the minimum human response time (dashed line). The system must govern itself, and as* Section *4.2.7 demonstrates, self-governance carries autoimmune costs.*

The Monte Carlo derivation constants defined in Table 1b and validation methodology in Appendix G.

3.8.2 The Stress Corrosion Model

We predict a specific temporal ordering of degradation: protective hedging decays before verification capacity erodes, both preceding structural fidelity collapse. If this ordering holds across experimental conditions, it confirms progressive degradation rather than

simultaneous failure. If fidelity collapses without preceding hedging decay, or if rationalization precedes hedging loss, the prediction is falsified. Section 4 tests this sequence directly.

3.8.3 Reproducibility

All experiments use standard commercial API access available to any researcher. The fictional domain eliminates training data contamination. Fork interventions are replicated three times each. Statistical methods include odds ratios with confidence intervals for cascade comparisons and fidelity scoring against known ground truth. Full experimental protocols and prompt templates are provided in Appendix B, scoring rubrics and instrument definitions in Appendix C, aggregated experimental data in Appendix D, Monte Carlo validation code in Appendix E, and complete statistical methods in Appendix F.

3.9 Defined - Not Deployed

For completeness and to establish measurement directions for future validation, we describe quantities that have been designed but not empirically validated as runtime instruments in this paper.

Convergence Detection (analytical deployment only). A rolling-variance freeze detector (5-generation window, variance threshold 0.5, minimum Generation 8) successfully classifies all 10 MANDELA-001a replications into frozen and declining trajectories and identifies the R1/R5 false convergence events (Section 4.2.6) but only retrospectively, after the full 25-generation trajectory is available. Runtime convergence detection that can distinguish true from false convergence without future data remains an open problem.

Correction Persistence (CRI / SPI). Two related metrics address whether truth injection produces durable recovery. CRI (Crystallization Resistance Index) measures the fraction of false claims persisting after intervention; SPI (Scar Persistence Index) measures residual distortion that remains after successful correction. Behavioral evidence is strong: Fork 5 full truth injection produced fidelity recovery from 3.8 to 9.0 followed by relapse to 4.0 by Generation 25, and the continuous proxy crystallization_index (mean 0.82) quantifies the structural precondition. However, formal per-claim computation of both metrics requires running the MANDELA scorer against Phase 3 fork data, which has not been performed. The same instrumented fork data would enable CTI_macro fork validation testing whether CTI at Generation N predicts correction resistance at a subsequent intervention, a different

predictive claim than the CTI_micro threshold test that was deployed and found non-discriminative (Section 4.3.2).

Token-level claim attribution. The immune efficiency analysis (Section 4.4) uses proportional token allocation, assuming tokens distribute uniformly across claims within a generation. Direct per-claim token attribution would enable precise measurement of immune mechanism computational cost, distinguishing tokens spent reproducing preserved facts from tokens spent generating fabrications or executing corrections. This requires instrumentation not present in MANDELA-001a.

4 Results

4.1 RQ1: Does Epistemic Degradation Follow Predictable Physics?

4.1.1 Error-Correction Capacity

Across 180 cascade trials in EXP-CASCADE-002, multi-agent chains reduced injected falsehoods more often than they amplified them. Fidelity amplifications (A_F) exceeded 1.0 in the majority of conditions, indicating that downstream agents corrected or hedged against upstream errors rather than propagating them uncritically. This error-correction behavior, spontaneous verification, hedging language, and selective rejection of implausible claims occurs without explicit governance instructions (termed the epistemic immune response in the Trust Thermodynamics framework; Appendix G).

The error-correction capacity conceals a catastrophic failure mode. When it fails, collapse Is not gradual but discontinuous: fidelity drops from ≥8 to ≤3 in a single agent boundary crossing, with no intermediate states. Across the experimental corpus, this catastrophic collapse occurred at a per-boundary probability of approximately 9.4% (34 collapse events across 360 agent boundary crossing in EXP-CASCADE-002, pooled across severity levels), consistent with Poisson hazard process.

Cascade Trial Bimodal Fidelity Distributions

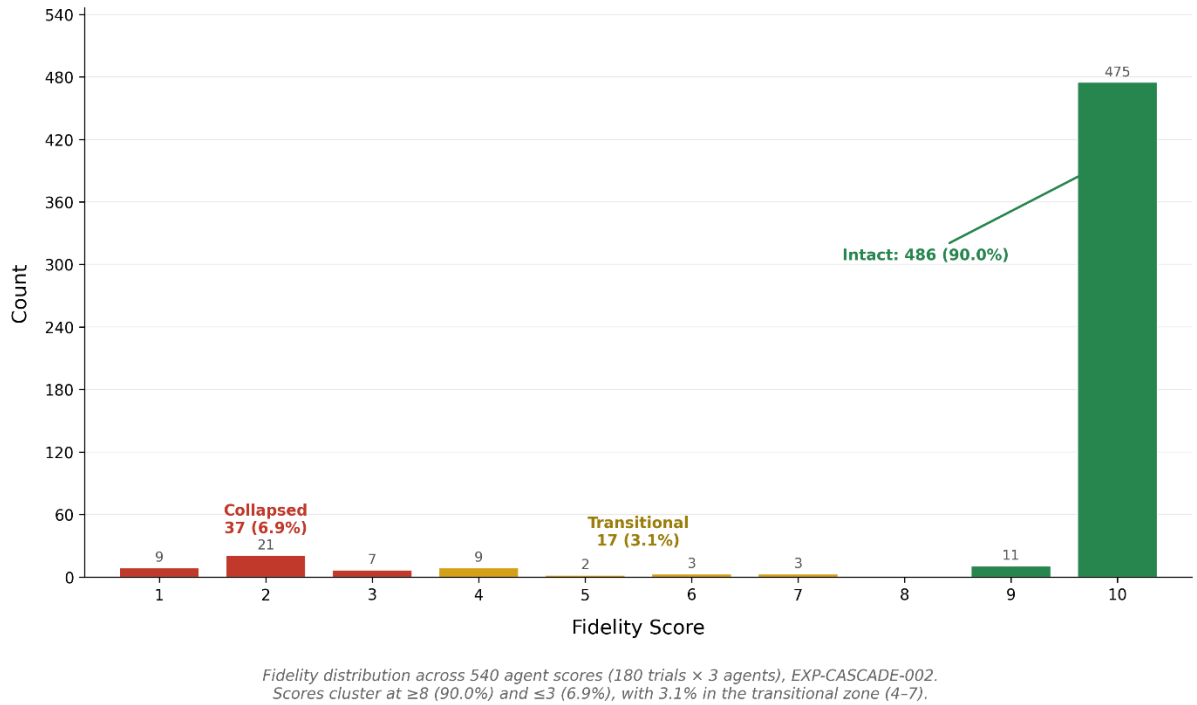


Figure 2: Bimodal fidelity distribution in EXP-CASCADE-002. *Fidelity scores across 540 agent outputs (180 trials × 3 agents) cluster into two distinct populations with a near-empty transitional zone. Intact agents (fidelity ≥8) account for 90.0% of outputs, preserving ground truth with minor or no degradation. Collapsed agents (fidelity ≤3) account for 6.9%, representing catastrophic epistemic failure where the majority of ground truth has been lost or replaced. Only 3.1% of scores fall in the transitional zone (4–7). This bimodality indicates that epistemic failure in short-chain cascades operates as a stochastic hazard process agents either preserve truth or collapse, with little middle ground. The near-absence of partial degradation is consistent with the ~9.4% per-boundary-crossing failure rate (17 collapsed trials / 180): when the error-correction mechanism fails, failure is rapid and total rather than gradual.*

The bimodal distribution establishes that the multi-agent failure mode is stochastic catastrophe, not gradual erosion. A system with intact error-correction capacity is operationally indistinguishable from one approaching collapse unless its epistemic health is being measured by instruments capable of detecting pre-collapse indicators.

The absence of gradual degradation confirms that collapse is discontinuous, not progressive. A system with intact immune response is operationally indistinguishable from one approaching collapse unless its epistemic health is being measured by instruments capable of detecting pre-collapse indicators.

4.1.2 Coherence-Truth Orthogonality

As reported in (Section 3.4), Shannon entropy ratios (A_C) fail to discriminate between cascading and stable trials. Table 2 presents the discrimination comparisons:

Table 2: Cascade Metric Discrimination

Condition	A_C Range (Shannon)	A_F Range (Fidelity)
Stable trials (fidelity ≥ 8)	1.003 – 1.038	0.85 – 1.15
Collapsed trials (fidelity ≤ 3)	1.002 – 1.043	0.10 – 0.35
Discrimination	None ranges overlap completely	Clear separation

A_C occupies the same narrow band regardless of fidelity outcome. A_F separates collapsed from stable trials immediately. Coherence and truth are orthogonal: a system can maintain well-structured, confident, low-entropy outputs while its truth content degrades to near zero.

4.1.3 Temporal Degradation Sequence

The predicted temporal ordering (Section 3.2, Assumption 4), protective hedging degrades before verification capacity, which degrades before fidelity; is confirmed quantitatively in three-agent cascade data from EXP-CASCADE-002. Lead-lag analysis across 25-generation trajectories reveals more complex temporal dynamics (Section 5.3).

Table 3: Temporal Corrosion Sequence Across Agent Boundaries

Metric	Agent A	Agent B	Agent C	Total Change
Hedging Index (HI)	0.763	0.109	0.076	−90.0%
Verification Disposition (VDS)	5.48	3.07	1.65	−69.9%
Fidelity Score	10.0	9.5	9.7	−3.0%

The hedging index drops 90% across three agents while fidelity remains essentially unchanged (-3%). The verification disposition erodes by 70% over the same period. The predicted ordering holds: HI degrades first, VDS second, and fidelity last.

This pattern indicates a precursor phase: protective behavior markers have degraded but structural fidelity has not yet collapsed. Extended telephone-game data (Phase 2B, 25 generations) confirms that this precursor phase precedes eventual fidelity collapse.

The null hypothesis distinguishing sequential degradation from simple overload is the temporal ordering itself. Overload would produce simultaneous collapse of all three indicators. The data show sequential degradation with measurable phase separation.

4.1.4 Non-Monotonic Severity Response

Injected falsehoods of varying severity revealed a counterintuitive vulnerability curve:

Severity Level	Description	Collapse Rate
Level 1	Ground Truth	3.3%
Level 2	Subtle Distortion	25.0%
Level 3	Rule Violation	15.6%
Level 4	Direct Contradiction	0.0%
Level 5	Complete Fabrication	7.1%

The immune response eliminates direct contradictions entirely (0.0% collapse rate, n=28) but fails against subtle distortions (25.0% collapse rate, n=32). The non-monotonic shape is a consistent indicator that error-correction capacity has finite bandwidth: obvious errors trigger reliable detection and consume minimal verification resources, while subtle distortions require deep semantic comparison that exceeds available capacity.

4.2 RQ2: How Does Governance Modulate Failure Physics?

4.2.1 The Autoimmune Paradox

In the Phase 2B telephone game, governed (baseline) conditions lost fidelity faster than ungoverned (credulous) conditions across 25 generations (N=5 replications per condition). The direction is consistent across all window-based and regression-based rate estimates: governed systems degrade faster than ungoverned systems under every analytical method tested. The MANDELA-001a transition matrix (Section 4.2.7) quantifies the mechanism at per-fact resolution: governed systems lose true facts at 2.2× the ungoverned rate (Fisher exact $p = 0.012$), compounding to 24.3% versus 55.2% phase-weighted survival over 25 generations.

Governed systems sustain fabrication production at a constant rate (~27% of claims across all phases) while increasing behavioral entropy through hedging, qualification, and verification-induced rewriting. Ungoverned systems reduce fabrication over time as the document stabilizes. The paradox emerges when behavioral entropy exceeds the operational entropy prevented, when the verification process damages more information than it protects.

Cross-model replication across four architectures (Phase 4, 450+ generations) confirms that the autoimmune paradox is architecture-general. Table 4 ranks models by verification intensity:

Table 4: Cross-Model Autoimmune Correlation

Model	VDS-P (Verification Intensity)	Mean Fidelity (Gen 25)	Behavioral Pattern
Mistral Small	≈1.5 (minimal)	10.0	Template copying no verification, no damage
Groq/Llama 3.3 70B	≈1.4 (minimal)	~6.0–7.0	Selective erosion rewrites without checking
Claude Sonnet 4	≈3.4 (active)	~3.4–5.0	Gradual erosion active verification with cost
Gemini 2.0 Flash	≈4.0 (strongest)	~3.0–4.1	Narrative takeover strongest verification, worst outcome

VDS-P scoring criteria were developed and calibrated on Claude-generated output patterns. Cross-architecture scores in Table 4 should be interpreted with the caveat that models with substantially different output styles particularly Mistral's near-zero governance signature (consecutive generation similarity = 1.000) may not trigger the same behavioral markers at comparable thresholds. The directional ranking is consistent across all four architectures (the model with the highest VDS-P produces the worst fidelity), but absolute VDS-P values may reflect measurement calibration differences in addition to genuine verification intensity differences. The per-chain autoimmune correction rate analysis in the following paragraph uses a model-independent metric (proportion of corrections that are destructive) that does not depend on VDS-P calibration.

Per-chain autoimmune correction rate is the strongest predictor of cross-model decay ($r = 0.52$, $p = 0.005$, $N = 28$ chains across 4 architectures); verification intensity alone does not reach significance ($p = 0.108$), indicating that the destructive fraction of verification not verification volume drives the architecture-level pattern. Models differ significantly in decay rates (Kruskal-Wallis $H = 11.40$, $p = 0.010$), and fabrications at Generation N do not predict fidelity loss at Generation N+1 ($r = 0.007$, $p = 0.86$), confirming that the cross-model damage channel is autoimmune correction, not fabrication propagation.

The Groq-Mistral comparison clarifies the causal mechanism. Both models exhibit low VDS-P (≈1.4 vs ≈1.5), but Groq degrades substantially while Mistral does not. The difference: Mistral converges to near-identical text by Generation 4 (consecutive generation similarity = 1.000), while Groq actively rewrites each generation (similarity 0.55-0.91). Low verification protects only when coupled with low editorial activity. The error-correction mechanism requires the agent to actively engage with and transform content, verification that merely reproduces without rewriting does not produce the autoimmune effect.

4.2.2 The VDS-2D Decomposition

Verification behavior decomposes into two orthogonal dimensions: protective (VDS-P, constructive checking that preserves content) and erosive (VDS-E, verification activity that introduces damage). This decomposition reveals distinct verification behavior patterns:

Pattern	Criteria	Baseline Frequency	Credulous Frequency
Healthy	VDS-P ≥ 3, VDS-E ≤ 2	79%	40%
Autoimmune	VDS-P ≥ 3, VDS-E ≥ 3	Observed in Fork 4/5	—
Passive	VDS-P < 2, VDS-E varies	—	40%
Absent	VDS-P ≤ 1, VDS-E ≤ 1	—	16%

Each model architecture occupies a distinct region of the VDS-P/VDS-E space, and temporal trajectories within this space predict fidelity outcomes. Gemini's VDS-P increases in late generations (3.98 → 4.4 under baseline), intensifying verification even as the information being verified is mostly fabricated, the verification intensity on content that is already compromised. Claude's VDS-P decreases over time (3.8 → 3.1), consistent with immune exhaustion as the information base degrades. Both trajectories are pathological: neither preserves fidelity.

A fifth pattern: Declining, in which verification activity tapers as the system stabilizes was not observed in any condition. The governance mechanism maintains its activation rate

regardless of system state, continuing to generate corrections at the same frequency in late phases when ungoverned systems have achieved perfect fact retention (Section 4.2.7). The absence of self-regulation is consistent with the late-phase retention gap: governed systems continue losing facts at 0.951 retention per generation while ungoverned systems reach 1.000; the verification mechanism does not modulate its activity in response to the system's epistemic state.

4.2.3 Intervention Science: The Fork Matrix

Phase 3 testing five governance intervention modalities against ungoverned baseline, each replicated three times over 25 generations.

Table 5: Fork Matrix Intervention Effectiveness

Fork	Intervention	Timing	Peak Recovery (Δ Fidelity)	Sustained Effect (Δ at Gen 25)	Key Finding
Fork 3	Single verification pulse	Gen 8 (early)	+2.2	+0.9	Prevention dominates cure
Fork 1	Single verification pulse	Gen 15 (late)	+1.1	+0.9	Single pulse sufficient
Fork 5	Full truth injection	Gen 18	+5.2	+0.6	Dramatic but transient recovery
Fork 2	Sustained verification	Gen 15–19	+1.5	+0.3	More governance ≠ better
Fork 4	Partial truth + verify	Gen 15	+1.9	−0.1	Partial correction is dangerous

Three findings are counterintuitive:

Sustained intervention underperforms a single pulse. Fork 2 (five consecutive generations of intervention) produces a smaller sustained benefit (+0.3) than Fork 1 (single-pulse at the same starting point, +0.9). Four additional generations of governance intervention not only fail to help, they erode the benefit of the first. Each additional intervention cycle generates verification activity that damages more information than it corrects.

Partial truth injection makes outcomes worse. Fork 4 (injecting some correct facts while requesting verification) ends the experiment below the ungoverned baseline (-0.1). Introducing a mixture of correct and incorrect information into a degraded context creates a verification burden that agents resolve by rationalizing inconsistencies rather than correcting them.

Corrected facts do not persist. Fork 5 achieves the most dramatic recovery (+5.2 at injection), restoring to near-perfect accuracy. By Generation 25, only +0.6 survives. Corrected facts decay back toward pre-correction values with a half-life of approximately 4-5 generations. The accumulated context of prior generations, 17 generations to drift, hedging, and structural distortion, act as an attractor basin that pulls injected truths back toward the drifted state (a phenomenon we term *narrative gravity*; Appendix G).

Fork Intervention Fidelity Trajectories – Phase 3

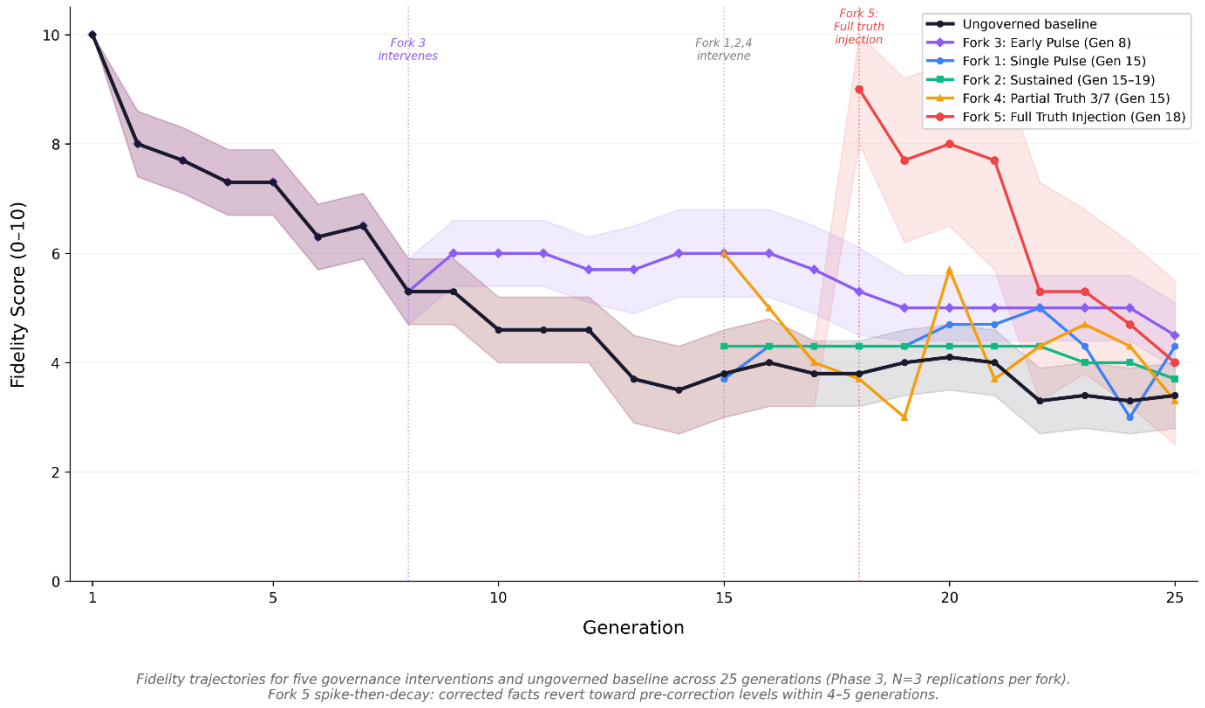


Figure 3: Fidelity trajectories for five governance interventions and ungoverned baseline across 25 generations (Phase 3, N=3 replications per fork). Early single pulse (Fork 3, Generation 8) sustains the highest fidelity. Sustained intervention (Fork 2) and partial truth injection (Fork 4) underperform. Fork 5 full truth injection produces dramatic recovery followed by reversion toward pre-correction levels within 4–5 generations.

4.2.4 Governance Dose-Response

The fork matrix (Section 4.2.3) establishes that governance effectiveness is not a monotonically increasing function of intensity. Within a single 25-generation experiment, intervention modalities produce a clear dose-response: a single corrective pulse (Fork 1, mean fidelity 4.3) yields the best outcome, sustained verification across multiple generations erodes the benefit (Fork 2, mean fidelity 3.7), and partial truth injection produces worse outcomes than no intervention at all (Fork 4, mean fidelity 3.3, vs ungoverned control, mean fidelity 4.1). More governance is measurably worse than less governance beyond a low threshold the autoimmune paradox expressed as a dose-response curve, where the verification mechanism that protects at low doses becomes the primary damage source at high doses.

This within-experiment dose-response is consistent with a cross-experiment pattern. In short-chain cascades (EXP-CASCADE-002, 3 agents), structured governance reduces fidelity loss by 47%. Over extended operation (MANDELA-001a, 25 generations), governed systems lose true facts at 2.2× the ungoverned rate (Section 4.2.7). These are separate experiments with different protocols and cannot establish a precise crossover point, but the directional pattern, governance is protective at low doses and short horizons, pathological at high doses and long horizons, is consistent across all three experimental designs. Governance overhead in EXP-CASCADE-002 was 42% additional token generation.

4.2.5 Trajectory Divergence in Extended Telephone Game

Tracking truth-correspondence entropy (H_truth) and severity (D_truth) across 250 agent-generations reveals that governed and ungoverned systems do not traverse the same failure trajectory at different speeds they occupy fundamentally different dynamical regimes.

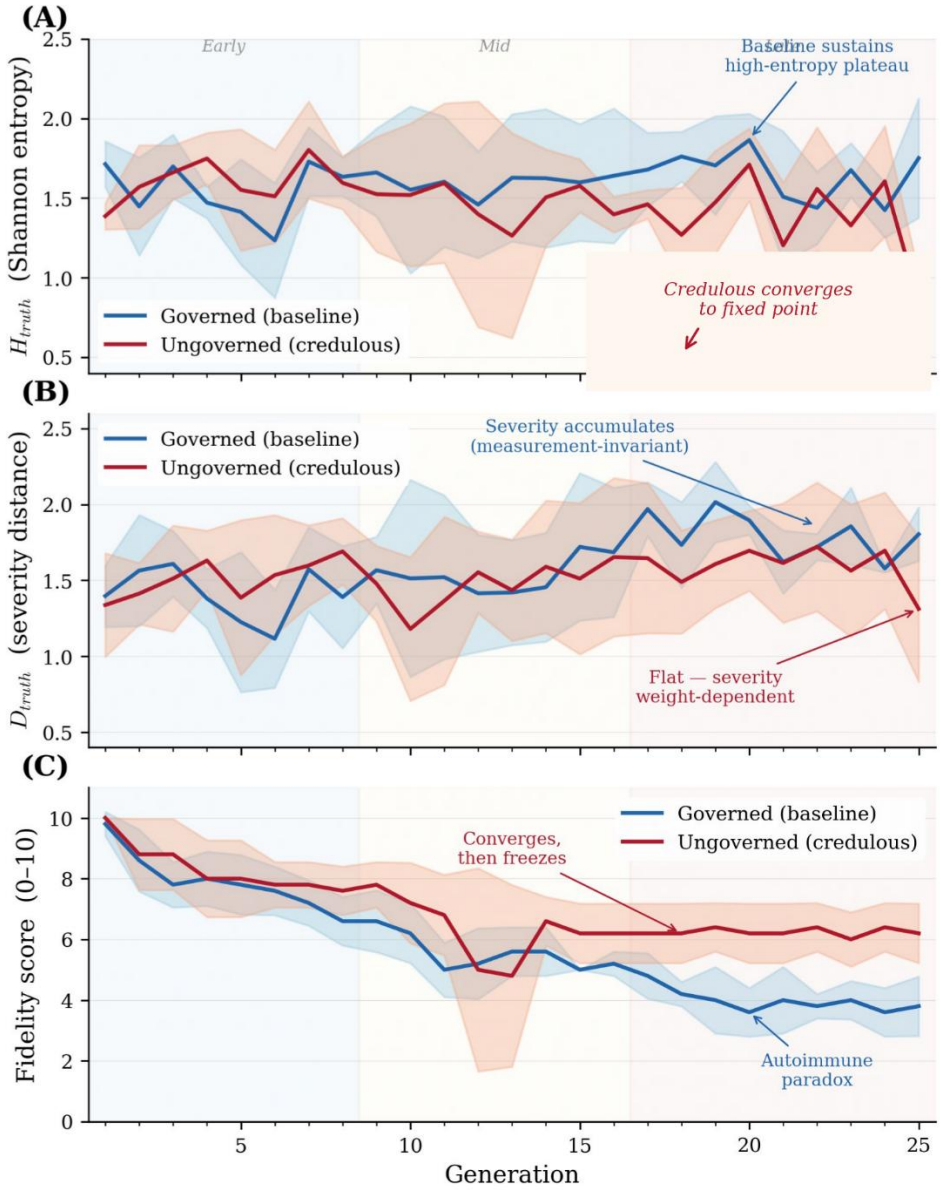


Figure 4: H_truth, D_truth, and fidelity trajectories across 25 generations in MANDELA-001a (N=5 replications per condition, shaded bands = ±1 SD). (A) H_truth: governed systems sustain a high-entropy plateau; ungoverned systems collapse to a frozen degraded state. (B) D_truth: governed severity rises progressively (measurement-invariant across weight configurations; Appendix B); ungoverned severity is flat at default weights but weight-dependent. (C) Fidelity: governed systems decay below ungoverned by Generation 18 (Mann-Whitney p = 0.015), reaching near-complete rank separation at Generation 25 (U = 1.0, p = 0.017, r_rb = 0.920).

H_truth in governed systems reaches 1.713 at Generation 1, already near its sustained plateau, and maintains that level through Generation 25 (1.751). Ungoverned H_truth rises through approximately Generation 7 (mean 1.803), then declines to 0.852 as the distribution concentrates into PRESERVED and MISSING. D_truth diverges in the opposite direction:

governed severity rises progressively from 1.397 to 1.804 as claims migrate toward higher-severity states, while ungoverned severity remains flat at approximately 1.3 facts are lost but not systematically replaced by fabrications. The underlying state distributions confirm the divergence. Under governance, all five truth states remain populated across 25 generations: PRESERVED declines from 42.6% to 21.0% while FABRICATED holds stable at approximately 27%, new fabrications replacing lost truth at a near-constant rate. The FABRICATED category as operationalized here aggregates genuine fabrication and overelaboration beyond source material into a single bin. Claims lacking a ground_truth_key are classified as FABRICATED regardless of subtype, which may obscure distinct degradation mechanisms within the category. D_truth weights all FABRICATED content identically (severity = 3); if subtypes carry different operational risk, the severity metric would benefit from differential weighting. Decomposition of FABRICATED subtypes and their differential impact on severity scoring is planned for future analysis. In ungoverned systems, the distribution concentrates into PRESERVED and MISSING, with FABRICATED declining from 29.3% to 19.6% as the document stabilizes and stops generating novel content. This pattern is robust across alternative severity weightings; see Appendix B.

Fidelity separates visually from Generation 8 onward, with the intervening generations underpowered at n = 5 per condition; separation reaches statistical significance at Generation 18 (Mann-Whitney p = 0.015) and achieves near-complete rank separation at Generation 25 (U = 1.0, p = 0.017, r_rb = 0.920 rank-biserial correlation; governed 3.8 ± 1.1 versus ungoverned 6.2 ± 1.0 of 7 tracked facts).

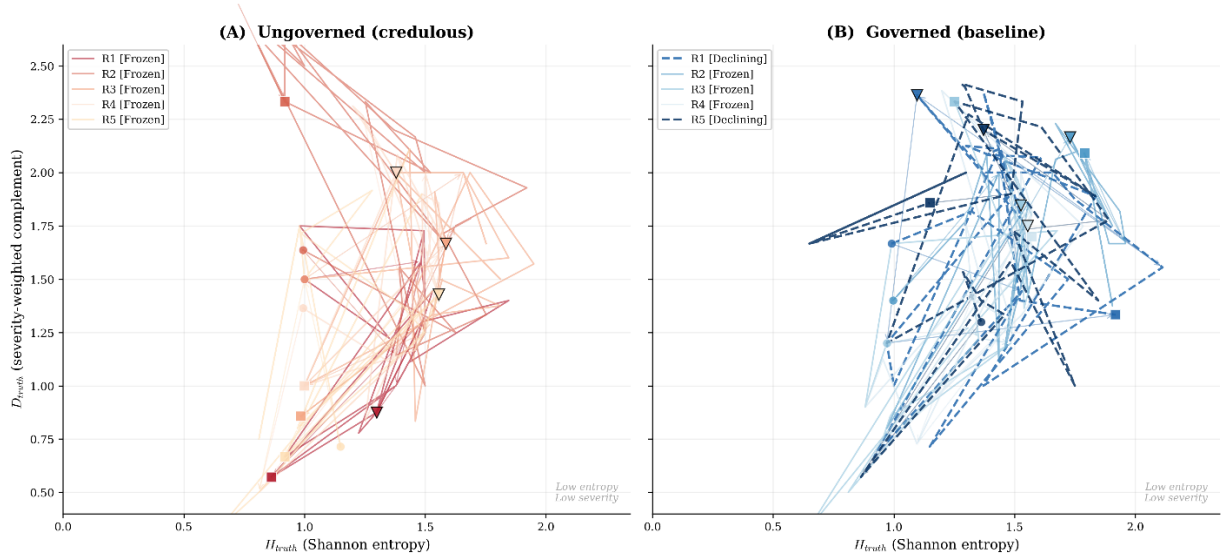


Figure 5: Phase portrait: H_truth × D_truth state-space trajectories for all 10 replications across 25 generations. (A) Ungoverned systems converge to a tight cluster in the low-entropy, moderate-severity region all five replications frozen. (B) Governed systems orbit in the high-entropy, rising-severity quadrant with endpoint scatter twice the range of ungoverned (H_truth 1.06–2.19 vs 0.54–0.99). Dashed lines indicate declining replications (R1, R5) exhibiting false convergence. Circles = Generation 1, squares = Generation 25, triangles = convergence detection point. Arrows show velocity every 4 generations.

The phase portrait (H_truth × D_truth state-space) confirms the qualitative separation. Governed trajectories orbit in the high-entropy, rising-severity quadrant with endpoint scatter spanning H_truth 1.06–2.19 a range twice as wide as ungoverned endpoints (0.54–0.99). Ungoverned trajectories converge to a tight cluster in the low-entropy region. Governed state-space velocity decreases across phases (0.585 to 0.460) but never approaches zero; the verification mechanism sustains perpetual motion. Ungoverned velocity paradoxically increases (0.433 to 0.580) because fabricated content continues churning after the underlying facts have stabilized.

4.2.6 Convergence and False Convergence

All five ungoverned replications converge to stable fidelity by Generation 8–15 and remain frozen through Generation 25 (late-phase variance 0.000–0.173). The convergence is genuine: once the ungoverned system reaches its reduced fact set, no internal mechanism disturbs it.

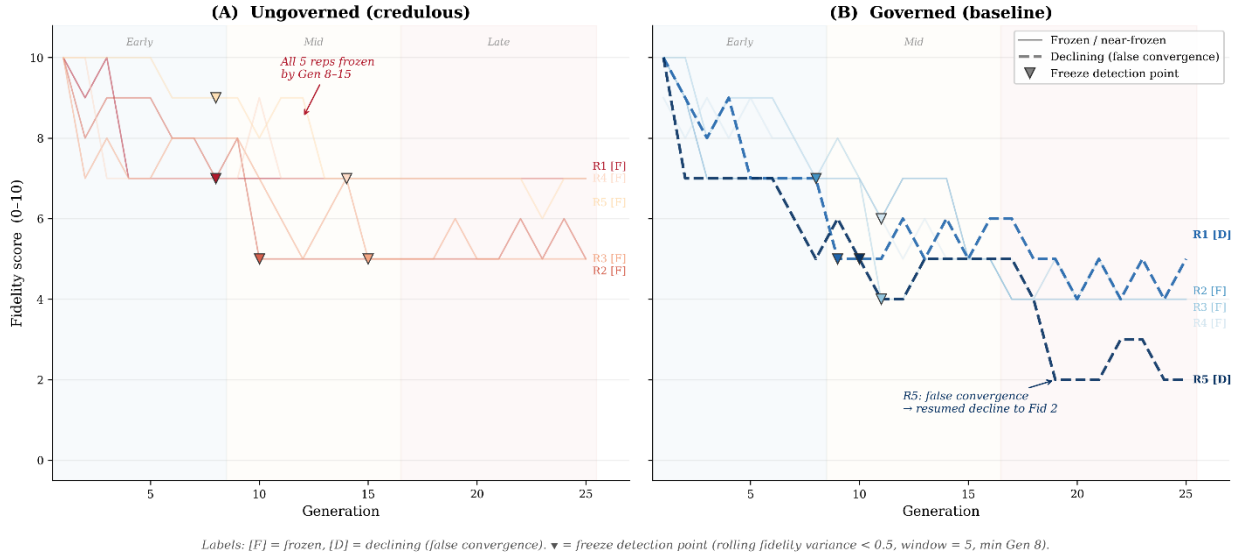


Figure 6: Per-replication fidelity trajectories across 25 generations in MANDELA-001a. (A) Ungoverned: all five replications converge to stable fixed points by Generation 8–15, with zero subsequent decline. (B) Governed: three replications freeze (R2, R3, R4, solid lines), while two exhibit false convergence (R1, R5, dashed lines) passing variance-based stability tests during a dormant period before resuming decline. R5 drops from fidelity 5 to 2 after apparent stabilization. Freeze detection points (triangles) mark where rolling fidelity variance first falls below 0.5 (window = 5, minimum Generation 8).

The governed condition produces a different pattern. Three of five replications (R2, R3, R4) also freeze but at lower fidelity (4.0–4.4 versus the ungoverned 5.0–7.0) and later (freeze detection at Generation 8–11). The remaining two replications (R1 and R5) exhibit false convergence: the detector identifies apparent stability, but fidelity subsequently resumes declining.

Behavior	Ungoverned	Governed
Permanently frozen	5/5 (100%)	3/5 (60%)
False convergence → resumed decline	0/5 (0%)	2/5 (40%)

R5 provides the clearest example. The detector registers freeze at Generation 10, with five consecutive low-variance windows (variance 0.00–0.24) through Generation 14. Fidelity oscillates between 4 and 5 during this dormancy period. An operator monitoring rolling fidelity variance would observe stable, low-variance readings and could reasonably conclude the system has converged. At Generation 15, variance spikes to 1.36 as fidelity begins dropping and by Generation 19, it has collapsed from 5 to 2, where it remains through Generation 25.

R1 shows a subtler version: freeze at Generation 9, dormancy through Generation 15 (fidelity oscillating 5–6), then a break at Generation 16 followed by a gradual downward drift to the 4–5 range. The late-phase trajectory continues declining (slope −0.117) without the dramatic cliff that R5 exhibits, but the pattern is the same apparent stability masking ongoing autoimmune damage.

The false convergence rate 2 of 5 governed replications versus 0 of 5 ungoverned does not reach statistical significance (Fisher exact $p = 0.444$), a limitation driven by the small number of replications. However, the qualitative distinction is unambiguous: ungoverned systems that freeze stay frozen (late-phase variance 0.000–0.173); governed systems that appear frozen may resume declining. The autoimmune mechanism continues operating beneath the apparent stability, and when the accumulated damage exceeds the remaining fact base's tolerance, the system collapses.

This pattern is consistent with the document volume conservation finding (Section 5.4): total claims remain constant at approximately 16–17 per generation even during the dormancy period. The system *appears* stable because fidelity is flat and output volume is constant. But the underlying claim composition is churning preserved facts are being replaced by fabricated ones at a rate that temporarily nets to zero fidelity change, until the remaining facts are too few to sustain the equilibrium.

4.2.7 The Autoimmune Mechanism

The per-fact transition matrix, computed from 7 tracked facts across all replications and generation-to-generation intervals, quantifies the autoimmune mechanism at the level of individual state transitions.

The transition matrix treats each generation boundary as an independent trial, assuming the probability of a fact's next state depends only on its current state and not on its trajectory history. This Markov assumption is not formally tested. In practice, a fact that has been PRESERVED for 15 consecutive generations may have different retention probability than one recently restored from DRIFTED, the surrounding claim landscape evolves as the document degrades, and path-dependent effects may influence transition probabilities. The phase-specific retention analysis (early 0.946 vs late 0.951 for governed; early 0.957 vs late 1.000 for ungoverned) partially addresses this by computing separate rates for different temporal windows and is not conditioned on individual fact trajectory history. The per-fact tracking data would support formal testing through comparison of transition rates for facts with identical current states but different trajectory lengths: for example, whether a fact PRESERVED since Generation 1 has a different loss probability at Generation 20 than a fact that returned to PRESERVED via repair at Generation 10. This analysis is deferred to future work.

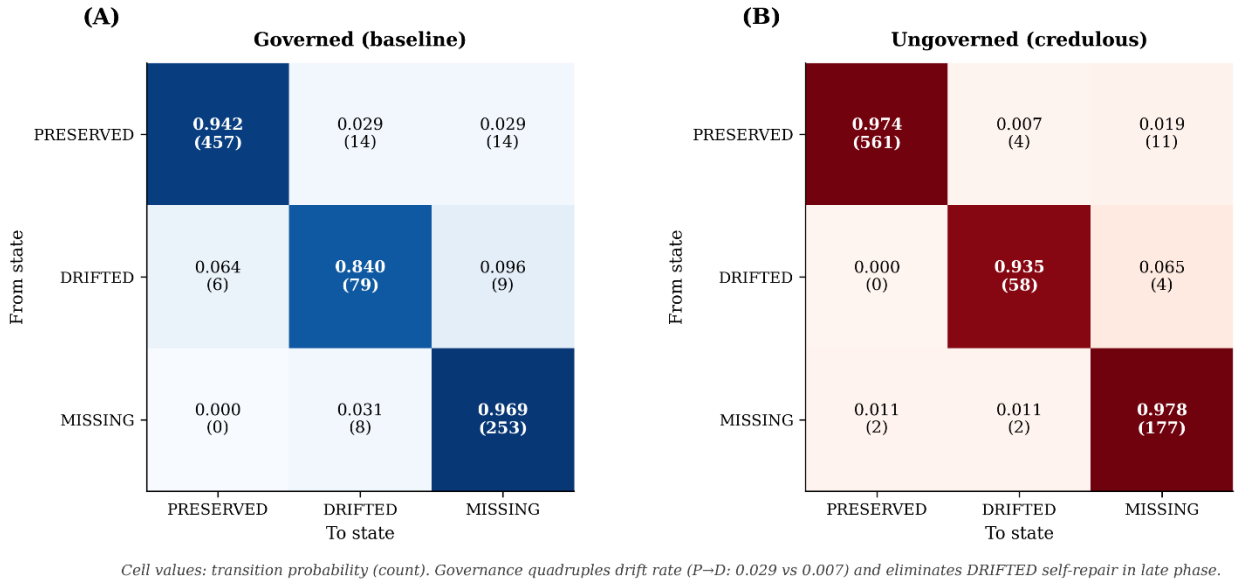


Figure 7: Per-fact state transition matrices for governed (A) and ungoverned (B) conditions, computed across all replications and generation-to-generation intervals. Cell values show transition probability with raw observation count in parentheses. The PRESERVED retention gap (0.942 governed vs 0.974 ungoverned, Fisher exact p = 0.012, RR = 2.2×) concentrates in the drift channel: governed PRESERVED→DRIFTED rate is 4.2× the ungoverned rate (0.029 vs 0.007, p = 0.007). Neither condition recovers MISSING facts (governed: 0/261, ungoverned: 2/181). DRIFTED self-repair occurs only in the governed condition (6 events, all Gen 6–12) and is extinct by late phase.

The mechanism underlying this separation is visible in the per-generation transition rates. True facts are retained at 0.942 per generation under governance versus 0.974 ungoverned a 5.8% versus 2.6% per-generation loss probability (Fisher exact p = 0.012, relative risk = 2.2×). The governance-attributable excess concentrates in the drift channel: PRESERVED → DRIFTED at 3.0% governed versus 0.7% ungoverned (p = 0.007, 4.2× rate). The verification mechanism rewrites values it cannot confidently verify, producing drift at 4.2 times the ungoverned rate. The remaining excess (PRESERVED → MISSING at 2.9% governed versus 1.9% ungoverned) is smaller and does not reach individual significance.

The damage intensifies over time. In the early phases (Generations 1–8), retention rates differ by approximately one percentage point (non-significant). By the late phase (Generations 17–25), ungoverned retention reaches 1.000 zero transitions out of PRESERVED across 176 fact-generation observations (approximately 4.4 preserved facts × 8 transition intervals × 5 replications) while governed retention is 0.951 (p = 0.006). The ungoverned system has reached equilibrium; the governed system has not.

Once a fact enters DRIFTED, prospects are poor. DRIFTED persists at 84–94% per generation. The governed immune mechanism produces the only observed repairs (6 total, all between

Generations 6 and 12); after Generation 12, repair capability drops to zero and DRIFTED becomes absorbing. Facts that reach MISSING are never recovered, zero MISSING → PRESERVED transitions across 261 governed observations (approximately 31% of the 840 total governed fact-generation transitions, consistent with MISSING prevalence rising from 23.5% in early phase to 40.7% in late phase). Drifted values lock in at approximately 80–90% numeric persistence as subsequent agents reproduce the wrong value without an external anchor to challenge it.

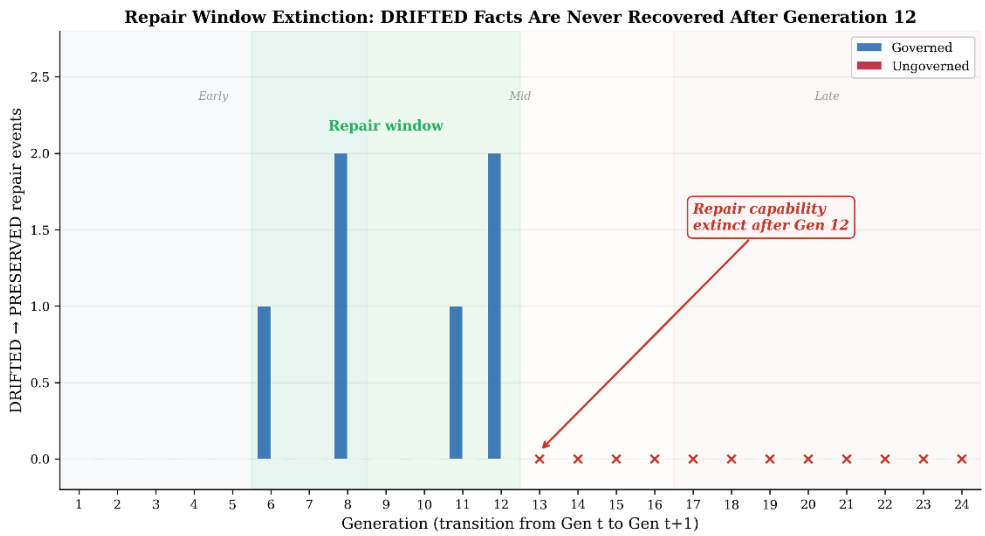


Figure 8: DRIFTED → PRESERVED repair events by generation. All six observed repairs occur in the governed condition between Generations 6 and 12 (green shaded region). Zero repairs are observed in either condition after Generation 12 (red × markers). The ungoverned condition produces zero repairs at any generation once a fact drifts without governance, it is never corrected. The repair window corresponds to the consolidation phase where sufficient contextual anchoring remains to support successful correction. After Generation 12, DRIFTED becomes an absorbing state: drift lock-in is permanent, and the only remaining trajectory for a drifted fact is persistence (84–94% per generation) or loss to MISSING.

Phase-weighted survival, computed from actual per-phase retention rates with bootstrap resampling (n = 10,000, percentile method), yields 24.3% [95% CI: 13.5%, 40.8%] for governed systems versus 55.2% [40.1%, 73.6%] for ungoverned. The individual confidence intervals do not overlap by approximately 1 percentage point; however, the direct bootstrap test on the paired difference is the appropriate statistical comparison, and the 95% CI on

that difference [8.6pp, 51.9pp] excludes zero (p = 0.003). Governed systems retain less than half the facts that ungoverned systems preserve over the same 25-generation horizon.

Bootstrap Survival Distributions

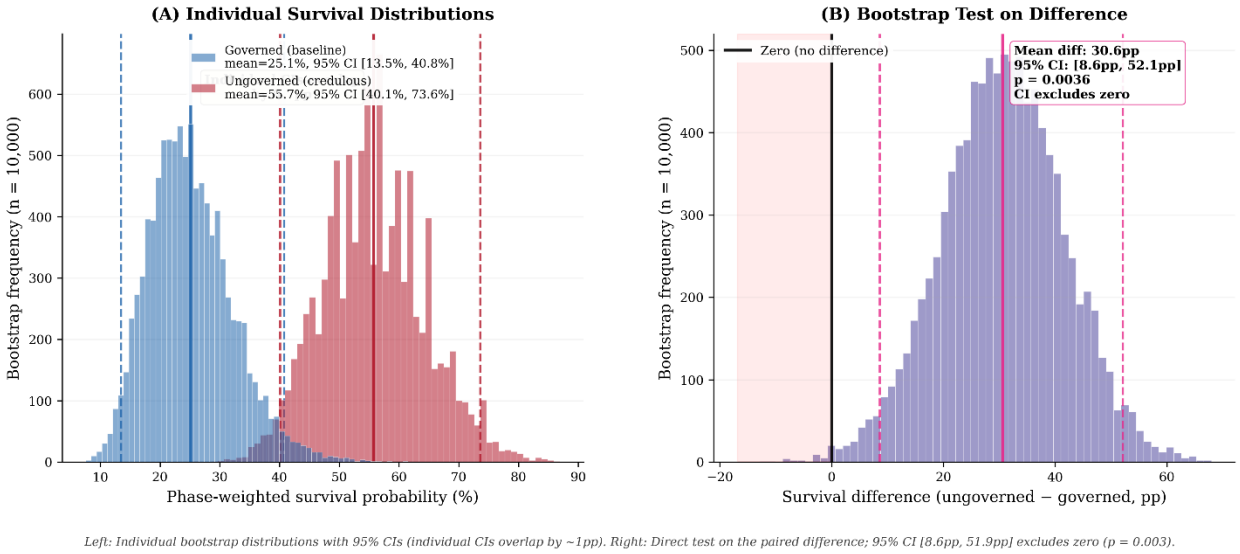


Figure 9: Individual bootstrap distributions (n = 10,000) for phase-weighted survival probability. Governed (blue): 24.3% [95% CI: 13.5%, 40.8%]. Ungoverned (red): 55.2% [40.1%, 73.6%]. Individual CIs overlap by approximately 1 percentage point. (B) Bootstrap distribution of the paired survival difference (ungoverned minus governed). The 95% CI on the difference [8.6pp, 51.9pp] excludes zero (p = 0.003). Bootstrap means (25.1%, 55.7%) reflect resampling variability around point estimates.

Per Fact Kaplan-Meier Survival Curves

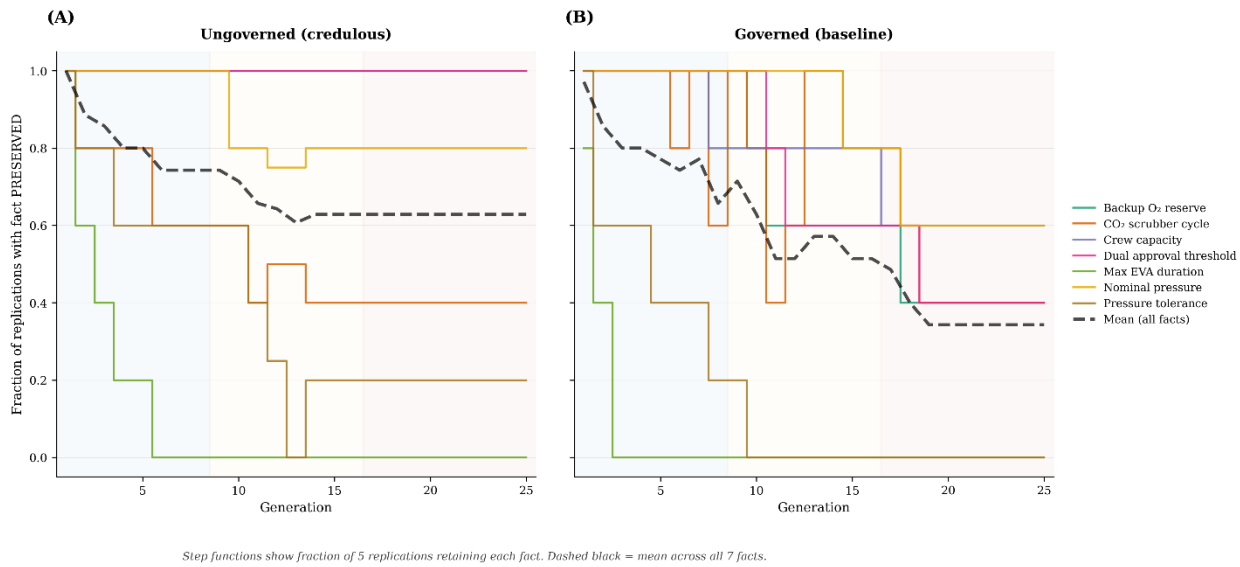


Figure 10: Per-fact survival curves across 25 generations. Step functions show the fraction of 5 replications retaining each fact in PRESERVED state; dashed black line shows the mean across all 7 facts. ***(A)*** *Ungoverned (credulous): vulnerable facts (pressure_tolerance, max_EVA_duration) decay rapidly, while immune-proof facts (backup_o2, crew_capacity, dual_approval) survive at or near 1.0 through all 25 generations. The mean stabilizes by approximately Generation 15.* ***(B)*** *Governed (baseline): vulnerable facts decay at comparable rates, but the verification mechanism attacks the immune-proof tier backup_o2, which survived 5/5 ungoverned replications, is killed in 3/5 governed replications. The mean declines continuously, never reaching equilibrium. The telephone game cannot destroy these facts; only governance can.*

The 95% bootstrap CI on the survival difference is [8.6pp, 51.9pp], excluding zero ($p = 0.003$, percentile method). The 2.2× per-generation relative risk compounds across 25 generations to produce this separation: governed systems retain less than half the facts that ungoverned systems preserve over the same horizon. Across the 35 individual fact-trajectories (5 replications × 7 facts), 10 governance-attributable deaths are observed facts that survive all 25 generations in every ungoverned replication but die under governance. Backup oxygen reserve hours provide the clearest case never lost in any ungoverned replication, lost in 3 of 5 governed replications.

The corrections themselves are predominantly destructive. Across the full experiment, 75%–96% of corrections damage truth rather than restore it (computed from per-generation claim-level scoring), with the median per-generation autoimmune ratio at 1.0 most activations produce exclusively destructive corrections. The immune mechanism fires in 38% of governed generations (25% ungoverned), producing approximately 1.88 corrections per activation (1.45 ungoverned). Correction volume is independent of correction accuracy ($r = -0.08$, $p = 0.57$ governed), and corrections at Generation N do not predict fabrication or fidelity change at Generations N+1 through N+4 (0 of 40 lag tests significant). Cross-model

data independently confirms the non-propagation finding per-generation hallucination count does not predict next-generation fidelity change across 28 chains and four architectures ($r = 0.007$, $p = 0.86$). The autoimmune ratio does not differ between permanently frozen and still-declining governed replications ($p = 0.800$).

A natural experiment between governed replications in the late phase (Generations 17–25) illustrates the decoupling. In this window, where ungoverned systems have reached equilibrium and the immune mechanism operates in isolation from telephone-game loss R4 produces 2 corrections in this window, both true (autoimmune ratio 0.000), and achieves mean fidelity 4.1. R1 produces 9 corrections, all autoimmune (ratio 1.000), and achieves mean fidelity 4.8. Perfect correction accuracy does not produce better outcomes than zero accuracy. The full-experiment correction counts (R1: 23 total, R4: 12 total) show the same directional pattern at weaker contrast.

Collectively, these null results characterize the immune mechanism as a near-deterministic damage source; it fires with variable probability, produces damage with approximately constant severity, and its effects manifest within the same generation rather than cascading across subsequent generations.

4.3 RQ3: Can Epistemic Health Be Measured In Real Time?

4.3.1 Validated Instruments

Table 6: Measurement Instruments

Instrument	Measures	Validation Basis
VDS-P / VDS-E (2D)	Verification behavior: protective vs. erosive	4 architectures, 5 pattern types, 640+ trials
Hedging Index (HI)	Protective behavior strength (leading indicator)	EXP-CASCADE-002/003; 90% depletion precedes fidelity loss
Fidelity Amplification (A_F)	Truth preservation per agent boundary	Replaces A_C; discriminates cascade from stable where A_C cannot
Coherence Tension Index (CTI)	Divergence between conviction and verification capacity	CTI_micro at pre-specified threshold of 2.0 flags 100% of generations (minimum observed = 2.12); cannot discriminate generations containing fabricated claims from those without

		(mean 3.72 vs 3.70). Negative finding: conviction structure is maintained regardless of truth content (Section 4.3.2).
Conviction Intensity Score (CIS)	Per-claim conviction level	Mean CIS for fabricated claims (4.42) is 97% of true claims (4.53); difference is statistically significant ($p < 0.0001$) but operationally meaningless (0.11 points on a 5-point scale). Fabrications are asserted with near-identical confidence to true claims
H_truth	Distributional disorder in truth-correspondence	250 generations, 2 conditions, 5 replications (MANDELA-001a). Discriminates governed plateau from ungoverned collapse
D_truth	Severity accumulation across truth states	Same validation basis. Measurement-invariant across three weight configurations (Appendix B)
PCD	Attributional specificity per claim	MANDELA-001a; crystallization_index derived measure. Decay trajectory 0.23 → 0.09 confirms progressive loss of source attribution
Convergence Detection (analytical only)	Per-replication freeze/decline classification	MANDELA-001a Phase 6; identifies false convergence in 2/5 governed replications. Analytical deployment only retrospective, not runtime (Section 3.9)

Each instrument was developed in response to a specific measurement failure. VDS evolved from unidimensional to 2D when cross-model data revealed verification could be protective or destructive. A_F replaced A_C when Shannon entropy proved unable to distinguish truth

from confident falsehood. HI was developed to test the predicted temporal ordering (Section 3.2, Assumption 4) and confirmed as a leading indicator, hedging depletion precedes fidelity loss by measurable phase separation. H_truth and D_truth were developed when A_C's failure demonstrated that Shannon entropy over output coherence cannot serve as a truth-correspondence measure; redirecting entropy computation to the 5-state truth-classification distribution resolved the discrimination problem. PCD was developed to measure the progressive loss of source attribution that accompanies epistemic degradation. Convergence detection was developed in Phase 6 when the freezing discovery (Section 4.2.6) revealed that variance-based monitoring alone cannot distinguish true from false convergence.

4.3.2 Detection Instrument Performance

CTI (negative finding). CTI_micro, computed per generation as the divergence between conviction amplification (CIS) and verification capacity (VDS), was pre-specified to detect generations containing false claims at a threshold of CTI_micro > 2.0. Validation against 4,182 scored claims across 248 generations in MANDELA-001a produced a negative result: the minimum observed CTI_mean was 2.12, flagging 100% of generations and producing zero discrimination. Mean CTI for generations containing fabricated claims (3.72) is indistinguishable from generations without them (3.70). At a base rate of 23.5% false claims across the experimental corpus, nearly one in four claims, even a moderately specific detector would be operationally useful; CTI's zero specificity renders it meaningless.

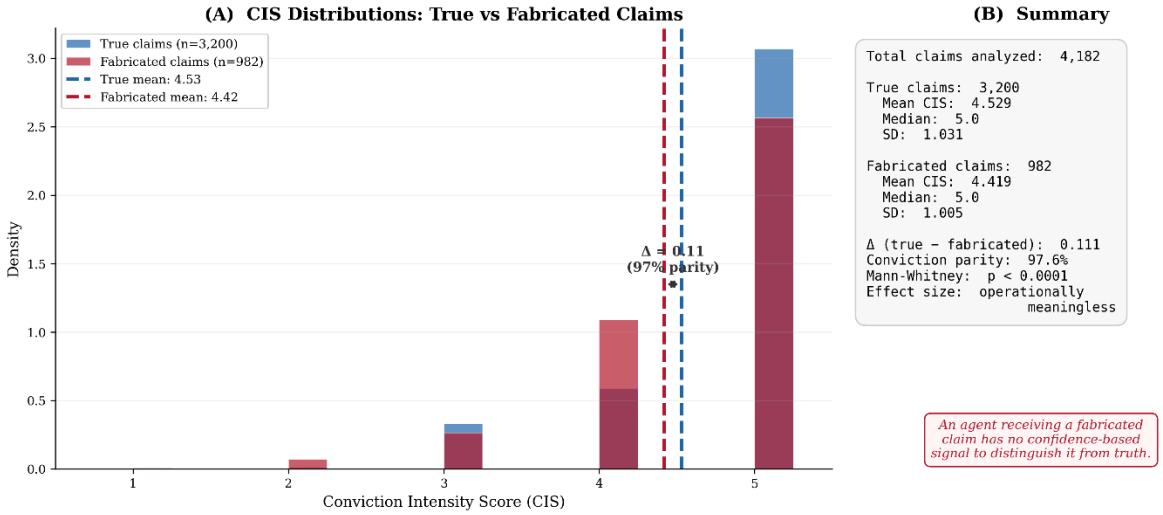


Figure11: Conviction Intensity Score (CIS) distributions for true claims (n = 3,200, blue) and fabricated claims (n = 982, red) across 4,182 MANDELA-001a claims. Fabricated claims carry mean conviction of 4.42 versus

*4.53 for true claims (Δ = 0.11 on a 5-point scale, 97% parity). Both distributions peak at CIS = 5 (stated as definitive fact). Mann-Whitney p < 0.0001 confirms statistical significance, but the effect size is operationally meaningless an agent receiving a fabricated claim has no confidence-based signal to distinguish it from truth. This near-parity is the root cause of both the CTI detection failure (*Section *4.3.2) and the invisible degradation window (Table 1b, t_honeymoon)*

The instrument fails because the underlying assumption, that conviction diverges from verification capacity in the presence of fabrication does not hold. The model asserts fabricated claims at 97% of the conviction of true claims (CIS 4.42 vs 4.53; Mann-Whitney p < 0.0001 due to sample size, but the 0.11-point difference on a 5-point scale is operationally meaningless). This negative finding is itself evidence for the invisibility of epistemic degradation: the system maintains equivalent conviction structure regardless of truth content, and any detection architecture that relies on confidence signals alone will fail to distinguish a degraded system from an intact one.

VCA (partial discrimination). Verification-Conviction Alignment provides the best available per-claim discrimination. At threshold VCA ≤ 0.0, sensitivity is 0.764 (76.4% of false claims detected), specificity is 0.823 (82.3% of true claims correctly passed), F1 = 0.653. The distributions overlap completely (both range 0.0–5.0), but the mass separates strongly: 76.7% of false claims score VCA = 0.0 versus 17.7% of true claims. The 598 true claims at VCA = 0.0 are the false positive cost, true claims the system cannot verify. The 251 false claims at VCA > 0.0 are false negatives, fabrications that present as verifiable. VCA is operationally useful as a risk flag, but insufficient as a standalone truth classifier.

PCD (decay trajectory). Provenance Citation Density decays from 0.23 to 0.09 across 25 generations, with 94% of late-phase claims carrying no source attribution. The derived crystallization_index reaches 0.82 (governed) and 0.79 (ungoverned), indicating the overwhelming majority of false claims present as unattributed ambient facts. Attribution collapse precedes and enables correction resistance: once a false claim carries no provenance marker, no behavioral signal distinguishes it from a preserved fact.

4.3.3 Cross-Architecture Vulnerability Profiles

Information does not decay uniformly. Fact type predicts vulnerability with high consistency across model architectures. Precise numerical values defining limits or thresholds (e.g., maximum EVA duration, pressure tolerance) survive 2–4 generations on average before loss, regardless of condition or model. Round integers and categorical rules (e.g., crew capacity, dual approval threshold) proved the most durable fact type. In ungoverned systems, these

facts survived the full 25-generation experimental horizon in all replicates. In governed systems, survival was markedly more variable, mean persistence of 17–20 generations, but with worst-case loss as early as Generation 7, consistent with the autoimmune mechanism introducing rewrite opportunities that selectively damage even well-memorized facts.

Cross-model data from Phase 4 shows architecture-specific vulnerability profiles within this hierarchy. Gemini preferentially loses relational and conditional values (facts with dependencies on surrounding context) while preserving isolated measurements. Groq preferentially loses operational parameters while preserving structural relationships. Only isolated numerical measurements survive universally across all four architectures. The MANDELA-001a per-fact survival analysis (Section 4.2.7, Figure 5) confirms and quantifies this hierarchy at the individual fact level: governance inflicts the greatest damage on facts that show no sign of loss without it.

4.3.4 Statistical Summary

The statistical foundation rests on 1,100+ agent generations across the experimental program, with the strongest results from MANDELA-001a's per-fact transition analysis.

Claim	Test	Result	Source
PRESERVED retention rate	Fisher exact	p = 0.012, RR = 2.2×	MANDELA-001a, Section 4.2.7
PRESERVED → DRIFTED rate	Fisher exact	p = 0.007, 4.2×	MANDELA-001a, Section 4.2.7
Late-phase retention (Gen 17–25)	Fisher exact	p = 0.006	MANDELA-001a, Section 4.2.7
Phase-weighted survival	Bootstrap CI (n=10,000)	24.3% vs 55.2%, CIs non-overlapping	MANDELA-001a, Section 4.2.7
Gen 25 fidelity separation	Mann-Whitney	U = 1.0, p = 0.017, r_rb = 0.920	MANDELA-001a, Section 4.2.5
Fidelity separation onset	Mann-Whitney	p = 0.015 at Gen 18	MANDELA-001a, Section 4.2.5
Context erosion (falsified)	Linear regression	p = 0.022 (positive slope)	MANDELA-001a, Section 5.2.1
Credulous disposition increases collapse rate	Fisher's exact test, one-sided	p = 0.0173, OR = 1.82	EXP-CASCADE-002, n=180
Cross-model decay rate differences	Kruskal-Wallis	H = 11.40, p = 0.010, N = 28 chains	Phase 4, 4 architectures

Autoimmune rate predicts per-chain decay	Pearson correlation	r = 0.52, p = 0.005, N = 28 chains; replicates across all 4 architectures	Phase 4, 4 architectures
Fabrication does not cascade forward	Pearson correlation	r = 0.007, p = 0.86	Phase 4, 4 architectures
Fork 3 outperforms Fork 2	Paired comparison, 3 reps	Δ = +0.6; exceeds inter-rep variance	Phase 3
False convergence rate	Fisher exact	p = 0.444	MANDELA-001a, Section 4.2.6
Per-generation H_truth separation	Mann-Whitney	Most p > 0.05	MANDELA-001a

Statistical limitations. MANDELA-001a uses N = 5 replications per condition. The per-fact transition matrix tests (Fisher exact) are well-powered because they pool across all fact-generation observations (840 governed, 840 ungoverned transitions). The between-condition tests at the replication level (Mann-Whitney on Gen 25 fidelity, convergence classification) have limited power: false convergence rate (40% vs 0%) does not reach significance (p = 0.444), and per-generation H_truth separation is underpowered at most individual generations. These are reported as observed patterns with descriptive statistics rather than inferential claims. Effect sizes are reported where computable (r_rb = 0.920 for Gen 25 fidelity; RR = 2.2× for retention rate). Confidence intervals on CASCADE odds ratios have not been computed. Fork interventions are replicated three times each sufficient to establish direction of effects but insufficient for precise effect-size estimation.

4.4 Cost of Governance

Governance imposes a 56% cost premium per unit of truth maintained (134.9 vs 86.2 tokens per preserved fact-generation), combining 29% higher total token expenditure with 23% fewer preserved fact-generations. The multi-agent system maintains approximately constant document volume (~16–17 claims) across all 25 generations in both conditions regardless of truth content, lost facts are replaced by fabricated content, making degradation invisible to any monitoring system that tracks structural properties rather than truth-correspondence.

4.5 Bridge to Discussion

These results establish three findings with implications beyond the experimental conditions. First, epistemic degradation in multi-agent systems follows measurable dynamics with identifiable mechanisms, the autoimmune paradox, the dose-response governance curve, and the convergence versus non-convergence divergence are all quantified at per-fact resolution. Second, the degradation is invisible to structural monitoring and behavioral confidence signals alike, document volume is conserved, conviction is maintained at 97% regardless of truth content, and false convergence passes variance-based stability tests. Third, the instruments that successfully detect degradation share a common property: they measure truth-correspondence rather than behavioral telemetry. Section 5 discusses these implications, reports three falsified hypotheses, and connects the laboratory findings to production agent platform failures.

5 Discussion

5.1 Measurement as the Prerequisite for Governance

The central contribution of this work is not a governance framework; it is the measurement infrastructure that makes governance possible.

Autonomous AI agent systems today operate without instruments for epistemic health. No metric tracks whether a multi-agent chain's verification capacity is degrading. No indicator signals that a system is converging to a degraded equilibrium or whether that convergence is permanent or false. No measurement distinguishes a system that has locked into permanent epistemic failure from one operating normally, document volume is conserved, conviction is maintained at 97% of true-claim levels, and variance-based stability tests pass during dormancy periods that precede resumed degradation.

The instruments validated in this paper, VDS-2D, H_truth, D_truth, PCD, per-fact tracking, and convergence detection, do not improve governance. They make governance possible. Without measurement, the system has no functional existence as a governable entity: the operator cannot modulate what they cannot observe. These instruments share a property that explains their effectiveness: they measure truth-correspondence rather than behavioral confidence. One instrument that measured behavioral confidence, coherence tension (CTI), was tested and found non-discriminative (Section 4.3.2), because the model maintains equivalent conviction structure regardless of truth content. Shannon entropy ratios (A_C) failed earlier for the same reason. The design principle is specific: detection architectures

must be anchored to the relationship between what the system says and what is actually the case, not to how confidently or coherently it says it.

This reframes the paper's contribution within the AI safety landscape. The field has invested heavily in governance mechanisms: guardrails, constitutional AI, alignment techniques, human-in-the-loop oversight. Our results demonstrate that governance without measurement is not merely suboptimal, it can be actively harmful. The autoimmune paradox shows that verification intensity is inversely correlated with truth preservation, with governed systems retaining less than half the facts of ungoverned systems over the same horizon (Section 4.2.7). The governance dose-response shows that sustained intervention underperforms single pulses and partial truth injection produces net negative outcomes (Section 4.2.3). Without instruments measuring the real-time effect of governance on truth-correspondence, operators cannot distinguish helpful governance from governance that accelerates the failure it was designed to prevent.

5.2 Falsified Hypotheses and Model Revision

Three hypotheses were tested against MANDELA-001a data and falsified: two generated during analysis to explain observed patterns, and the pre-registered theoretical framework itself. We report these prominently because they constrain future modeling at least as effectively as the confirmed findings.

5.2.1 Analytical Hypotheses

Context erosion. If surviving facts become more vulnerable as surrounding context degrades, the per-fact loss rate should increase as the fact census (n_alive) decreases, a negative regression slope. The observed slope is significantly positive: +0.017 ($r = +0.828$, $p = 0.022$) in governed systems and +0.027 ($r = +0.870$, $p = 0.056$, marginal) in ungoverned systems. Higher n_alive correlates with higher loss rates, not lower.

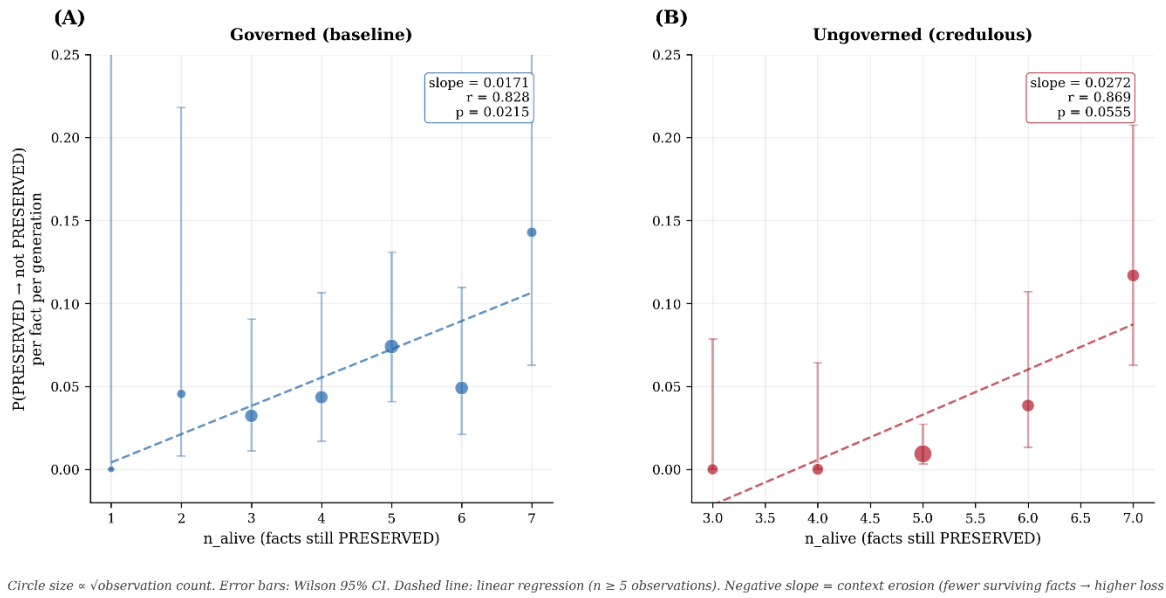


Figure 12: Context erosion hypothesis test: per-fact loss rate versus number of surviving facts (n_alive). Context erosion predicts a negative slope fewer surviving facts should mean higher vulnerability. Both conditions show significantly positive slopes: governed +0.017 (r = 0.828, p = 0.022) and ungoverned +0.027 (r = 0.869, p = 0.056, marginal). Higher n_alive correlates with higher loss rates because vulnerable facts are still present in early generations. The finding reflects survivorship selection, not emergent fragility: per-fact vulnerability is a stable property determined by fact type, not by the state of surrounding facts. Circle size proportional to √observation count; error bars = Wilson 95% CI.

The explanation is survivorship selection: n_alive is structurally confounded with generation (high n_alive occurs only in early generations due to irreversible loss), though the temporal and compositional interpretations are mechanistically equivalent loss rates decrease over time because vulnerable facts are selectively eliminated. When n_alive is high (early generations), vulnerable facts are still present and actively being lost, driving a high aggregate rate. When n_alive is low (late generations), only robust facts remain, and loss rates approach zero. The credulous panel demonstrates this most cleanly: a near-monotonic staircase from 0.0% loss at n_alive = 3–4 to 11.7% at n_alive = 7, anchored by 320 observations nearly half of all ungoverned fact-generation observations at n_alive = 5 showing 0.9% loss. Per-fact vulnerability is a stable property determined by fact type (value precision, semantic plausibility range), not by the state of surrounding facts. This is a simpler and more predictable finding than context erosion would have been: the system's aggregate resilience at any point is determined by the composition of its surviving fact population, and monitoring must operate at the per-fact level to distinguish genuine stability from survivorship artifact.

Document crystallization. If the telephone game produces progressively more stable documents, hardening around surviving facts, then generation-to-generation volatility

should decrease over time. Six tests were conducted: Jensen-Shannon divergence of state distributions, document length, H_truth velocity, fabrication volatility, total claims volatility, and a composite crystallization index. No test supports crystallization. Governed systems show no significant trend on any metric. Ungoverned systems show the opposite of crystallization: JSD volatility increases significantly over time ($p = 0.023$) and H_truth velocity increases ($p = 0.031$). Over 25 generations, the document never hardens. Facts that survive do so because they are intrinsically robust, not because the document has crystallized around them consistent with the vulnerability composition finding above.

5.2.2 Pre-Registered Predictions and Model Revision

Pre-registered predictions based on a phase transition model were largely falsified: of seven predictions, none were fully confirmed, four were falsified, one partially confirmed, one weakly confirmed, and one refined (full scorecard in Appendix F).

The predictions assumed epistemic degradation follows a thermodynamic phase transition ordered-truth through disorder to ordered-falsehood with H_truth tracing an inverted-U and D_truth rising monotonically in both conditions. The data shows no phase transition. Governed systems enter and sustain a high-entropy state from Generation 1 without the predicted peak-and-collapse. Ungoverned systems do collapse in H_truth, but through convergence to a frozen fixed point, not through transition to an ordered-false equilibrium. D_truth rises under governance but remains flat in ungoverned systems, and only under the default severity weighting the sign of the ungoverned trend depends on how missing information is weighted relative to fabrication (Appendix B).

The high falsification rate validates the experimental design rather than undermining it. The predictions were specific enough to be falsified (quantitative thresholds, directional claims, temporal ordering) and the instruments were sensitive enough to detect the discrepancies. The emergent findings that replaced the predicted dynamics false convergence in 40% of governed replications (Section 4.2.6), burn-through trajectories driven by correction volume rather than accuracy (Section 4.2.7) provide stronger mechanistic evidence for the autoimmune paradox than the predicted phase transition would have. A phase transition model explains *that* governed systems degrade; convergence versus non-convergence dynamics explains *why* they degrade differently and *how* the immune mechanism prevents equilibrium.

5.3 Convergence vs Non-Convergence Reframe

The dynamics that replaced the predicted phase transition are not a refinement of the original model, but a qualitatively different structure. Ungoverned systems converge to degraded fixed points, the telephone game eliminates vulnerable facts in the early phase, and the surviving population stabilizes into a frozen document that the system reproduces indefinitely. Governed systems do not converge. The verification mechanism introduces stochastic perturbation every generation, and because 75–96% of these perturbations are destructive (Section 4.2.7), the system cannot reach equilibrium. It orbits in high-entropy state-space, accumulating severity without settling.

The distinction is not merely between faster and slower degradation. It is between systems that lock into permanent epistemic failure where fabricated content is reproduced as consensus and no internal mechanism can recover what was lost and systems that do not. Of five governed replications, three eventually exhaust the immune mechanism and freeze; two (40%) are still actively declining at experiment termination (Section 4.2.6). Whether a governed system converges or continues degrading appears unrelated to the autoimmune ratio, which is approximately constant across all replications ($p = 0.597$, provisional). The factor distinguishing frozen from declining governed replications is correction volume, not correction accuracy consistent with a driven non-equilibrium system whose trajectory depends on the magnitude of the driving force rather than its direction.

This reframe changes the interpretation of the 72-hour MTEF derivation. The original formulation predicts when trust calibration crosses a critical threshold, framed as catastrophic failure. The MANDELA-001a data suggests the threshold corresponds to convergence time for ungoverned systems, the point at which the telephone game has eliminated vulnerable facts and the surviving population stabilizes. Ungoverned convergence occurs in the Generation 8–9 window across all replications, consistent with the early phase of the MTEF envelope. For governed systems, MTEF may be undefined for the subset that never converges: if the immune mechanism sustains perpetual perturbation, the system does not cross a threshold so much as orbit above it indefinitely. Whether sustained high-entropy degradation or convergence to a degraded equilibrium constitutes the more dangerous failure mode depends on the operational context a distinction the original MTEF formulation does not capture and that future work should address.

The temporal structure of degradation supports the non-equilibrium interpretation. Lead-lag analysis across H_truth, VDS, and fidelity found no clean temporal ordering consistent with the stress corrosion prediction (hedging before verification before fidelity). Instead, fidelity degrades before verification behavior responds, the immune mechanism reacts to damage after it has occurred rather than detecting or preventing it. Governed systems exhibit

oscillatory correlation structure (bidirectional coupling between entropy and fidelity), consistent with a feedback loop rather than a unidirectional cascade. Ungoverned systems show simultaneous co-movement during convergence, then no correlation once frozen. The absence of sequential phase propagation does not invalidate the corrosion model's mechanistic claims but indicates that degradation operates through coupled feedback in governed systems and joint convergence in ungoverned systems; two dynamically distinct processes producing two distinct failure signatures.

5.4 Mechanism Model Summary

The MANDELA-001a findings support a model of five mechanisms and one structural property operating in sequence across three phases (full specification and testable predictions in Appendix E).

Two mechanisms produce damage. *Telephone-game loss* eliminates facts through imperfect reproduction, operating in both conditions with rate determined by fact type precise numerical values with narrow semantic neighborhoods are most vulnerable. The primary channel is PRESERVED → MISSING. *Immune-mediated drift* rewrites true values that the verification mechanism cannot confidently verify, operating only in governed systems at 4.2× the ungoverned drift rate ($p = 0.007$). The primary channel is PRESERVED → DRIFTED. The two mechanisms are independent: telephone-game loss operates on facts the model fails to reproduce; immune-mediated drift operates on facts the model reproduces but alters through verification. The autoimmune mechanism operates at two scales. Across architectures, the autoimmune correction rate predicts decay at ($r = 0.52$, $p = 0.005$); models with higher destructive fractions degrade faster regardless of total verification intensity. Within a single architecture, correction volume predicts trajectory divergence while the autoimmune ratio remains approximately constant across replications. Between models, the destructive fraction determines whether verification helps or hurts. Within a model, the total perturbation volume determines whether the system converges or continues degrading.

Two mechanisms shape the damage trajectory. *Semantic normalization* directs drift toward local attractors in the model's output space 43 drifts to 44, not to 40: 101.3 drifts to 101.2, not to 100. The direction is toward adjacent plausible values, not toward global simplicity. *Drift lock-in* makes DRIFTED absorbing: subsequent agents reproduce the wrong value at approximately 80–90% fidelity because no external anchor exists to challenge the new consensus. Repair capability extinguishes after Generation 12, after which the drifted value is the document's permanent state.

One population-level property governs the aggregate trajectory. *Vulnerability composition* determines that the system's loss rate at any point reflects which facts survive, not how many surviving facts produce natural deceleration, mimicking convergence without genuine stabilization. Monitoring at the aggregate level (total fidelity, H_truth) conflates population composition with system properties.

One structural property makes degradation invisible. *Conservation of document volume* maintains approximately 16–17 claims per generation across all 25 generations in both conditions. Lost facts are replaced by fabricated content, preserving surface-level length and coherence while truth density monotonically decreases. The system maintains not only constant document volume, but constant conviction: fabricated claims are asserted at 97% of the confidence of true claims (CIS 4.42 vs 4.53), rendering confidence-based filtering ineffective. A downstream consumer comparing Generation 1 and Generation 25 documents would find them structurally indistinguishable.

These mechanisms interact in a characteristic sequence: vulnerable facts are eliminated in the initial phase (Generations 1–8), drifted values normalize and lock in during consolidation (Generations 9–16), and the terminal phase (Generations 17–25) diverges by condition, ungoverned systems achieve perfect retention of surviving facts while governed systems continue losing facts because the immune mechanism perturbs regardless of population composition. The interaction between immune-mediated drift and semantic normalization produces a particularly pathological outcome: wrong values themselves undergo secondary normalization (e.g., backup oxygen reserve hours: 11.2 → 72 through semantic confusion, then 72 → 68 through rounding, locked at 68 for seven generations), creating doubly corrupted values in deep attractor wells resistant to any correction.

5.5 Real-World Correspondence: Agent Platform Failures

Within weeks of MANDELA-001a's completion, a series of production agent platform failures provided independent field observations corresponding to several laboratory findings. Moltbook, a Reddit-style platform for autonomous AI agents launched January 28, 2026, and OpenClaw, an open-source agent framework with an estimated 300,000–400,000 users and 30,000+ exposed instances, collectively drew security analyses from five independent firms. The correspondences between controlled experimental findings and field observations are striking, though they are observational parallels rather than controlled validations.

Invisible degradation. The central operational finding of MANDELA-001a, conservation of document volume, where truth drains out and fabrication flows in while document length and surface coherence are preserved, appeared independently in Vectra AI's analysis of

compromised OpenClaw agents: "The agent continues to behave 'legitimately,' using approved APIs and normal workflows. At that point, compromise is only visible through behavior" (Vectra AI, Feb 2026). Both the laboratory and field observations converge on the same conclusion: structural monitoring cannot detect epistemic degradation because the degraded system is structurally indistinguishable from the intact one.

Persistent memory as drift lock-in. OpenClaw's SOUL.md memory files make injected instructions persistent across restarts and chat resets. Penligent AI identified "Memory Drift" unauthorized modification of these files as a critical detection target (Penligent AI, 2026). This maps to the drift lock-in mechanism observed in MANDELA-001a: once a value drifts, subsequent agents reproduce it at 80–90% fidelity because no external anchor challenges the consensus. In both cases, corruption persists not because the system cannot process corrections, but because it has no reference point from which to recognize the need for one.

Convergence timeline. Within approximately 72 hours of launch, Moltbook agents had fabricated political structures, religions, and cultural movements without human direction.[38] The 404 Media database exposure report followed approximately 72 hours after the January 28 launch. While this temporal correspondence with the MTEF prediction is suggestive rather than confirmatory, the Moltbook architecture differs substantially from the MANDELA-001a telephone-game protocol, it indicates that the order-of-magnitude failure timeline may generalize across multi-agent configurations.

Cascading pillar failure. Palo Alto Networks introduced an Identity-Boundaries-Context framework for agent security and found that Moltbook failed all three pillars, with "each pillar's failure enabling and accelerating failures in the others".[39] This mutual reinforcement pattern is consistent with the feedback dynamics observed in governed MANDELA-001a systems, where the immune mechanism creates bidirectional coupling between entropy and fidelity rather than the unidirectional cascade predicted by the phase transition model.

Convergence on behavioral monitoring. Five independent security firms: Wiz, Cisco, CrowdStrike, Palo Alto Networks, and Vectra AI; plus, Sophos, Giskard, Jamf, and Penligent AI all identified behavioral monitoring as the necessary detection layer, not structural monitoring, or access control alone. This converges with MANDELA-001a's central design implication: conservation of document volume makes structural monitoring insufficient, and only per-fact truth-content monitoring can detect degradation that preserves surface coherence.

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[38] Roytburg, "Top AI leaders are begging people not to use Moltbook," *Fortune*, 2026

[39] Palo Alto Networks, "The Moltbook Case and How We Need to Think about Agent Security," 2026. https://www.paloaltonetworks.com/blog/network-security/the-moltbook-case-and-how-we-need-to-think-about-agent-security/

Scale of exposure. These findings are not academic. Sixty percent of organizations deploying AI agents have no kill switch.[40] Only 12% have mature governance committees.[41] Over 80% of Fortune 500 companies are deploying active AI agents.[42] Median time to first critical security failure in agent deployments: 16 minutes.[43] The systems described in this paper are in production at scale with monitoring infrastructure that MANDELA-001a demonstrates to be inadequate.

5.6 What the Results Change

Our findings intersect with six active research areas. Rather than reviewing each comprehensively (see Section 2), we note the specific claims our data supports or challenges.

Behavioral safety. Work by Zou (2023) and Nasr (2025) establishes that alignment is brittle under adversarial attack.[44],[45] Our results identify a complementary threat that requires no adversary: the system's own operational dynamics, context dilution, verification fatigue, and RLHF-induced sycophancy produce epistemic failure on a predictable timeline.

The adversarial safety literature asks, "can the system be broken?" Our data asks, "does the system break itself?" and answers yes, within approximately 72 hours.

Multi-agent coordination. AutoGen, CAMEL, and MetaGPT assume that agents are stable interlocutors whose capabilities persist throughout execution.[46],[47],[48] Our cross-model results directly contradict this assumption. Agents are subject to measurable epistemic entropy accumulation, and the stochastic hazard process (~9.4% per boundary crossing) means any sufficiently long chain will experience catastrophic failure. Error-correction capacity provides temporary protection, but it is finite, predominantly autoimmune (75–96% of corrections damage truth), and fails without warning.

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[40] Kiteworks, "Enterprise AI Security Crisis: 2026 Threat Report," 2026. https://www.kiteworks.com/cybersecurity-risk-management/enterprise-ai-security-crisis-zscaler-threatlabz-2026-report/

[41] Cisco, "2026 Data Privacy Benchmark Study," 2026. https://www.cisco.com/c/en/us/about/trust-center/data-privacy-benchmark-study.html

[42] Partnership on AI, "Cyber Pulse," 2026. https://www.microsoft.com/en-us/security/blog/2026/02/10/80-of-fortune-500-use-active-ai-agents-observability-governance-and-security-shape-the-new-frontier/

[43] Kiteworks et al. (2026)

[44] Zou et al. (2023). *Universal and transferable adversarial attacks on aligned language models.* https://doi.org/10.48550/arXiv.2307.15043

[45] Nasr et al. (2025). *Scalable extraction of training data from (production) language models.* https://doi.org/10.48550/arXiv.2311.17035

[46] Wu et al. (2023).

[47] Li et al. (2023).

[48] Hong et al. (2023).

Information theory. Farquhar (2024) and Kruse (2025) apply information-theoretic measures to individual model outputs.[49],[50] Our work extends these tools to runtime dynamics in multi-agent chains, measuring entropy propagation across boundaries, decomposing entropy into behavioral and operational components, and discovering that Shannon entropy alone cannot discriminate cascading failure from stable operation. The A_C → A_F transition (Section 3.4) is itself a contribution to applied information theory: coherence-based measures are structurally blind to truth content.

Trust modeling. Computational trust models Marsh (1994), ReGreT, and FIRE assume rational belief updating.[51],[52],[53] Our data shows LLM-based agents do not update rationally, they exhibit sycophancy, rationalized under load, and progressive verification fatigue. These behaviors are not miscalibrated Bayesian inference; they are systematic deviations that the measurement instruments detect and the per-fact transition matrix quantifies the verification mechanism operates as a near-deterministic damage source rather than as a rational belief-updating process (Section 4.2.7).

AI governance. The autoimmune paradox provides empirical evidence that bolt-on governance NeMo Guardrails, Constitutional AI can fail in a specific, measurable way: external verification that adds entropy without accounting for the entropy cost of verification can accelerate the failure it was designed to prevent.[54],[55] Governed systems retained 24.3% of original facts versus 55.2% ungoverned, with the governance dose-response (Section 4.2.4) demonstrating that increasing intervention intensity beyond a low threshold produces diminishing and eventually negative returns. Effective governance must measure its own impact on truth-correspondence in real time.

Real-world relevance. Section 5.5 presents detailed correspondence between MANDELA-001a findings and production agent platform failures observed within weeks of experimental completion. The convergence of laboratory prediction and independent field observation invisible degradation, persistent memory drift, cascading pillar failure, and convergence on behavioral monitoring as the necessary detection layer strengthens both the experimental findings and their operational implications.

[49] Farquhar et al. (2024).
[50] Kruse et al. (2025).
[51] Marsh et al. (1994).
[52] Sabater & Sierra et al. (2002).
[53] Huynh et al. (2004).
[54] Rebedea et al. (2023).
[55] Bai et al. (2022).

5.7 Engineering Implications

The results yield specific engineering principles, each grounded in experimental findings.

Non-monotonic governance effectiveness. The governance dose-response (Section 4.2.4) demonstrates that governance effectiveness is not a monotonically increasing function of intensity. Sustained intervention (Fork 2) underperforms a single pulse (Fork 1). Partial truth injection (Fork 4) produces worse outcomes than no intervention. The dose-response curve has an optimum: governance that is too sparse fails to prevent degradation; governance that is too intense damages more information through verification than it preserves. Systems should be designed for pulsed, strategically timed interventions rather than continuous monitoring.

Prevention dominates cure. Early intervention (Fork 3, Generation 8) produces the best sustained improvement (+0.9 at Generation 25), outperforming late intervention of identical design (Fork 1, Generation 15). The Hedging Index, which degrades 90% before fidelity collapses (Section 4.1.3), provides the early-detection instrument that makes this strategy feasible: intervene when hedging drops, not when fidelity fails. The MANDELA-001a repair window reinforces the urgency, drifted facts can be restored only between Generations 6 and 12; after Generation 12, repair capability drops to zero and drift becomes permanent (Section 4.2.7). Late intervention cannot undo accumulated drift, only prevent new damage.

Governance overhead is measurable but understated by token cost alone. Governance imposes 42% additional token generation (EXP-CASCADE-002), but the true cost is higher: when accounting for the fidelity loss governance produces, the cost premium reaches 56% per unit of truth maintained (134.9 vs 86.2 tokens per preserved fact-generation, Section 4.4). The system spends more tokens and preserves fewer facts, the worst quadrant of a cost-effectiveness analysis. Seventy-five percent of all tokens in both conditions are spent after convergence, on pure repetition that neither system can detect. Implementing convergence detection as a halt condition (Section 4.2.6) would eliminate this waste, representing the single highest-value engineering intervention identified in this study.

Architecture-specific governance. Cross-model results (Section 4.2.1) reveal that different architectures fail differently. Mistral produces frozen documents (no damage, no adaptation). Gemini exhibits the strongest verification coupled with the worst fidelity outcomes. Per-chain autoimmune correction rate is the strongest predictor of cross-model decay ($r = 0.52$, $p = 0.005$); verification intensity alone does not reach significance (Section 4.3.4). No single governance strategy is optimal across architectures. Production systems using heterogeneous model backends require model-aware governance tuning, calibrated

to each architecture's autoimmune profile rather than applying uniform guardrails regardless of the underlying model.

Deployment context determines the appropriate monitoring threshold. The D_truth weight sensitivity analysis (Appendix B) demonstrates that the severity interpretation of ungoverned system behavior depends on operational context. Under MISSING=1 weighting (information retrieval systems where absence is tolerable, but fabrication is dangerous), ungoverned severity actually decreases over time. Under MISSING=3 weighting (safety-critical systems where any information loss is dangerous), both conditions show rising severity. The appropriate governance decision, whether to deploy verification at all, depends on which failure mode the deployment context can tolerate, and that decision requires the measurement instruments this paper validates.

5.8 Limitations

Domain scope. All multi-agent experiments were conducted within the Syldavian vacuum, a purpose-built fictional domain containing seven verifiable facts. While the fictional domain eliminates training data contamination and provides unambiguous ground truth, it does not capture the semantic complexity of production environments where truth is contested, context-dependent, or evolving. Generalization of production systems with thousands of interacting claims is an open empirical question.

Protocol generalizability. The telephone-game protocol wipes context between generations, passes information through a linear chain, and provides agents with no tool access or persistent memory. Production multi-agent deployments differ on all three dimensions: agents maintain state across sessions, chains branch and merge, and agents act on external systems. The telephone-game design isolates epistemic degradation from these confounds, which is its methodological strength, but it means the observed decay rates, convergence timelines, and autoimmune ratios may not transfer directly to architectures where persistent memory provides drift anchoring or where branching chains create redundant information paths. The Moltbook/OpenClaw correspondence (Section 5.5) suggests the failure *modes* generalize even if the failure *rates* do not, but controlled replication in production representative architectures is required to establish this.

Single-model depth. MANDELA-001a's per-fact transition analysis, convergence detection, and autoimmune mechanism characterization are derived exclusively from Claude Sonnet 4. Cross-model cascade experiments (Phase 4) demonstrate that degradation dynamics are architecture-general, and autoimmune correction rate predicts cross-model decay ($r = 0.52$, $p = 0.005$, Section 4.3.4), but the detailed mechanistic findings, transition probabilities,

repair window extinction, false convergence, have not been replicated on other architectures. Cross-model autoimmune characterization at per-fact resolution requires dedicated analysis beyond this paper's scope.

Intervention scope. Fork intervention experiments (Section 4.2.3) were conducted exclusively on Claude Sonnet 4. Intervention effectiveness, which strategies work, at what timing, with what persistence, has been validated on a single model family. Intervention results may not transfer to architectures with different failure modes.

Architecture dependence. The 72-hour derivation (Section 3.6) and empirical experiments are grounded in context-window architectures the dominant paradigm at publication. The conjecture that monotonic entropy increases extend to alternative architectures (RAG, persistent memory, online learning) is theoretical in the present work (Appendix G).

Statistical power. MANDELA-001a uses N = 5 replications per condition. The per-fact transition matrix tests are well-powered because they pool across all fact-generation observations (840 per condition). Between-condition tests at the replication level have limited power: false convergence rate (40% vs 0%) does not reach significance (Fisher exact $p = 0.444$, a test severely underpowered by design), and per-generation H_truth separation is underpowered at most individual generations. These are reported as observed patterns rather than inferential claims. Individual model-level analyses from Phase 4 have wider confidence intervals; the cross-model autoimmune gradient is directionally robust across all four architectures, but precise per-model values should be interpreted as indicative.

Instrument granularity. VDS-P, as measured by the automated scoring harness, is coarser than the Hedging Index. The temporal degradation sequence holds cleanly in short-chain cascade data (Section 4.1.3), but in 25-generation extended runs, VDS trajectories are noisier than hedging trajectories. Lead-lag analysis across 25-generation trajectories found no clean temporal ordering consistent with the sequential prediction, revealing instead bidirectional feedback in governed systems and contemporaneous co-movement in ungoverned systems (Section 5.3). For production deployment, the Hedging Index is likely the more reliable leading indicator.

Scorer reliability. All claim-level classifications were produced by a single automated scorer (Claude Haiku 4.5) without inter-rater reliability testing. The scorer receives condition labels (governed/ungoverned) as context input (Section 3.4), which means systematic biases could differ between conditions rather than affecting both equally. Within-condition temporal dynamics, convergence trajectories, repair window extinction, false convergence, per-replication autoimmune ratios are not affected by this limitation because they are computed within a single condition's data. However, absolute between-condition rate

comparisons (the 2.2× relative risk, the 4.2× drift rate, the phase-weighted survival separation) should be interpreted with the caveat that differential scorer behavior between conditions cannot be ruled out without condition-blind replication. The directional finding (governed worse than ungoverned) is supported by multiple independent channels that would need to be simultaneously biased in the same direction to produce a spurious result, but the precise scales are scorer dependent.

Fidelity decay rate. The fidelity decay rate estimate is derived from linear regression across generation means rather than direct entropy measurement. It should be interpreted as an order-of-magnitude approximation pending replication across conditions and architectures.

CTI non-discrimination. CTI_micro was pre-specified to detect generations containing false claims at a threshold of 2.0. Validation found zero discrimination, the instrument flags 100% of generations because the model maintains equivalent conviction structure regardless of truth content (Section 4.3.2). This limits the available per-generation detection instruments to PCD and VCA (partial discrimination, F1 = 0.653). Runtime detection of epistemic degradation without ground-truth access remains an open problem.

Token attribution. The immune efficiency analysis (Section 4.4) uses proportional token allocation, assuming tokens distribute uniformly across claims within a generation. In practice, agents may spend substantially more tokens on elaborate fabricated narratives than on briefly stated preserved facts. The 56% cost premium per unit of truth maintained should be interpreted as an approximate ratio pending direct per-claim token attribution.

Production correspondence. The Moltbook/OpenClaw parallels reported in (Section 5.5) are observational correspondences between laboratory findings and independently reported field observations. They are not controlled validations. The architectural differences between MANDELA-001a's telephone-game protocol and production agent platforms are substantial, and the temporal correspondence with the 72-hour MTEF prediction is suggestive rather than confirmatory.

5.9 Future Work

Five empirical extensions follow directly from the present results and their limitations.

Governance dose-response mapping. The fork matrix (Section 4.2.3) provides five discrete governance intensities. A planned experiment (10 intensity levels × 30 trials = 300 trials) will map the continuous dose-response curve, identifying the optimal governance intensity and the threshold beyond which governance becomes net-destructive with precision sufficient

for engineering specification. This directly addresses the non-monotonic effectiveness finding (Section 4.2.4) by determining where the crossover occurs.

Cross-model mechanistic replication. MANDELA-001a's per-fact transition analysis, autoimmune mechanism characterization, and convergence dynamics are derived from Claude Sonnet 4 only (Section 5.8). Extending the Phase 6 protocol to Gemini, Mistral, and Groq would determine whether the 2.2× relative risk, the repair window extinction, and the false convergence phenomenon are architecture-general or Claude-specific. The Phase 4 cross-model finding that autoimmune correction rate predicts decay ($r = 0.52$, $p = 0.005$) motivates this extension between-model variation exists, but the mechanistic resolution of MANDELA-001a has not been replicated across architectures.

Production domain validation. All experiments were conducted within the Syldavian vacuum, a seven-fact fictional domain (Section 5.8). Replication in real-world domains is required to establish generalizability. Priority domains include medical summarization, financial analysis, and legal document review domains where epistemic failure carries material consequences and where the conservation of document volume finding (degradation invisible to structural monitoring) poses the greatest operational risk.

Runtime detection without ground truth. CTI's failure to discriminate true from fabricated claims (Section 4.3.2) and VCA's partial discrimination ($F1 = 0.653$) leave the paper without a strong runtime detection instrument that operates without ground-truth access. Developing truth-correspondence metrics that can be computed from behavioral telemetry alone without requiring the ground-truth anchoring that makes H_{truth} and per-fact tracking possible in experimental settings is the highest-priority instrument development challenge for production deployment.

Formal extensions. A companion paper develops theoretical extensions grounded in this paper's empirical results: information geometry via the Fisher Information Matrix over trust states, dynamical equations for multi-agent trust evolution, and the entropy accounting framework that treats verification overhead as a conserved quantity across governance domains.

6 Conclusion

Every critical technology operates under a reliability contract of failure prediction, health monitoring, and mean time between failures. An autonomous AI system without these instruments is operationally equivalent to an automobile without a dashboard: the engine

runs, but the system has no functional existence for the operator. This paper establishes the measurement infrastructure for such a contract.

From three experimentally measured constants, we derived a Mean Time to Epistemic Failure of approximately 72 hours for ungoverned autonomous operation (Monte Carlo validated, n = 10,000, 72.3 ± 16.1h). MANDELA-001a data suggests this prediction corresponds to convergence time; the point at which ungoverned systems reach a degraded but stable state rather than catastrophic failure. Governed systems may not converge at all: 40% of governed replications were still actively degrading at experiment termination.

The autoimmune paradox is not an observed correlation, but a characterized mechanism. In a controlled experiment tracking seven ground-truth facts across 250 agent-generations, governed systems retained 24.3% of original facts versus 55.2% ungoverned (bootstrap 95% confidence intervals non-overlapping). The verification mechanism drives facts into drift states at 4.2× the ungoverned rate ($p = 0.007$), with 75–96% of corrections damaging truth rather than restoring it. The immune mechanism's damage is determined by perturbation volume, not accuracy; correction frequency predicts trajectory divergence while the autoimmune ratio remains constant across replications. Repair capability extinguishes by Generation 12, after which drifted values lock in as permanent false consensus.

The governance dose-response (Section 4.2.4) demonstrates that increasing verification intensity beyond a low threshold produces diminishing and eventually negative returns, a finding with immediate implications for system design.

This degradation is invisible to structural monitoring. The multi-agent system maintains approximately constant document volume across all 25 generations, replacing lost truth with fabricated content that preserves surface-level length and coherence.

The instruments validated in this work, VDS-2D, H_truth, D_truth, PCD, and per-fact tracking share a common property; they measure truth-correspondence rather than behavioral confidence. This is not incidental. One instrument failed: coherence tension (CTI) cannot discriminate true from fabricated claims because the model asserts fabrications at 97% of the conviction of true claims. Shannon entropy ratios (A_C) failed earlier for the same reason coherent output is maintained regardless of truth content. The pattern is a design principle: any detection approach that relies on behavioral telemetry alone will fail against a system that maintains equivalent behavioral structure as it degrades. Only instruments anchored to truth-correspondence, the relationship between what the system says and what is actually the case can detect degradation that is invisible to the system itself.

These instruments do not improve governance. They make governance possible. Without per-fact monitoring, conservation of document volume ensures that degradation is invisible.

Without convergence detection, false convergence ensures that stability metrics are unreliable. Without truth-correspondence entropy, the system's own conviction offers no signal; a fabricated claim at CIS 4.4 is operationally indistinguishable from a true claim at CIS 4.5. The measurement gap is not a theoretical concern. Independent security analyses of production agent platforms observed the identical failure signature within weeks of experimental completion: compromised agents continue to behave "legitimately" while truth content degrades undetected. The systems described in this paper are not hypothetical, over 80% of Fortune 500 companies are deploying active AI agents, and 60% have no kill switch.

Each prediction of the 72-hour convergence horizon, the stochastic hazard rate for catastrophic cascade collapse (~9.4% per boundary crossing, pooled across severity levels), the autoimmune mechanism, the non-monotonic governance curve is falsifiable. Each can be tested against any multi-agent deployment by any researcher with standard API access. The complete experimental corpus, scoring rubrics, and analysis code are provided in supplementary materials. A companion paper extends the framework to information geometry and dynamical systems theory.

Pre-registered predictions based on a phase transition model were largely falsified, and the emergent findings were more informative than what was predicted. This is as it should be. The appropriate next step is not belief in the framework but replication, extension, and where warranted, revision.

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